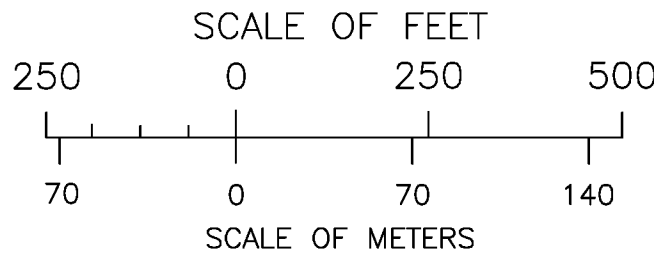
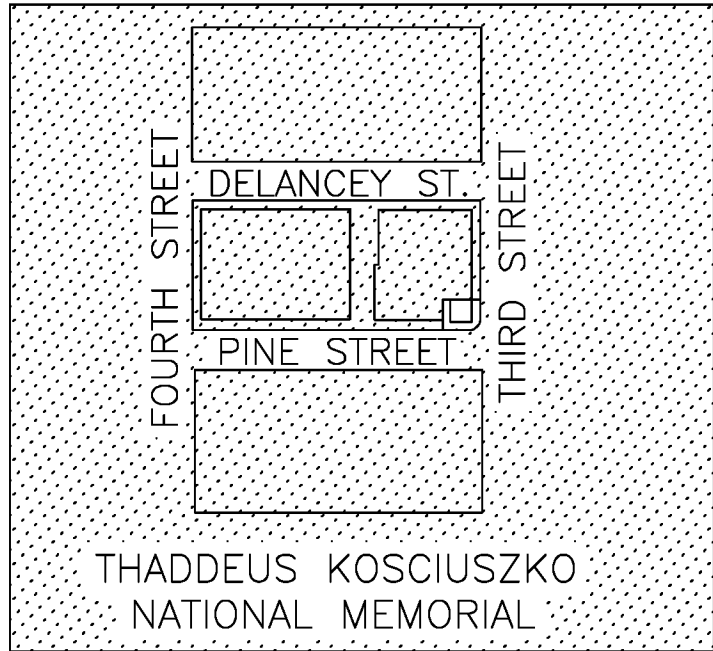
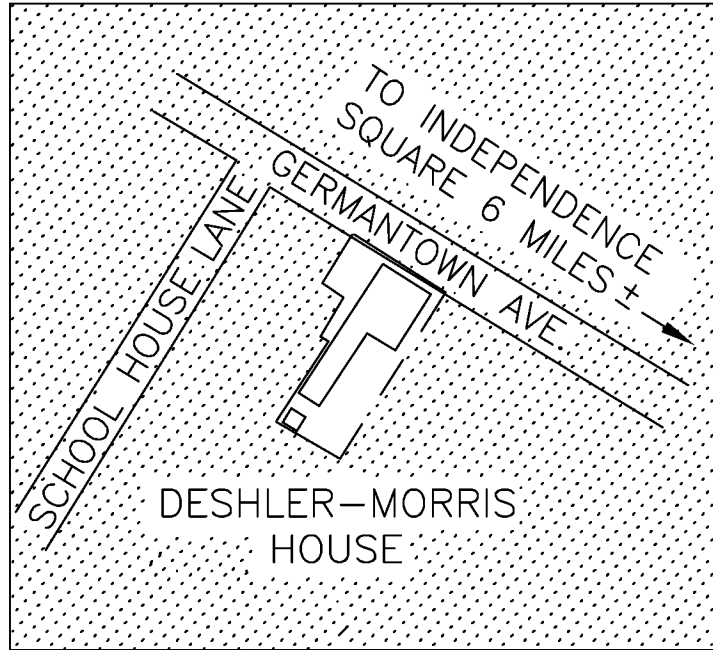
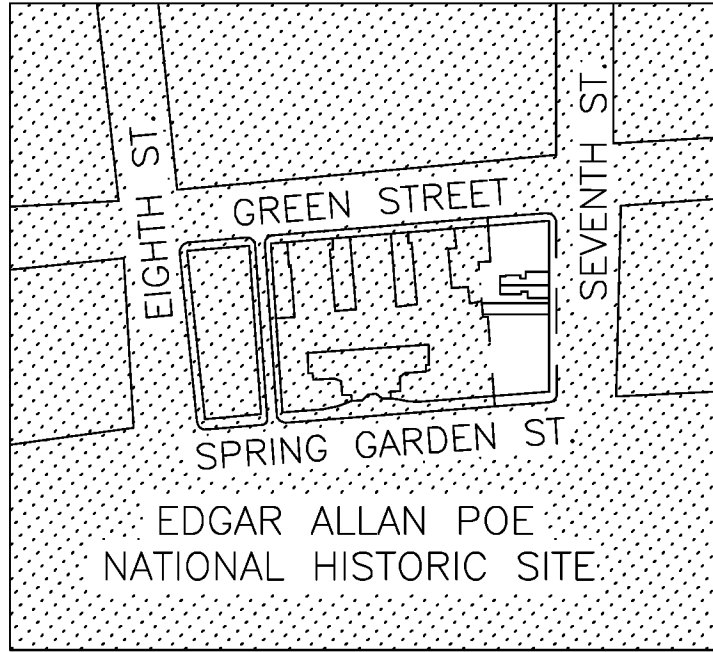
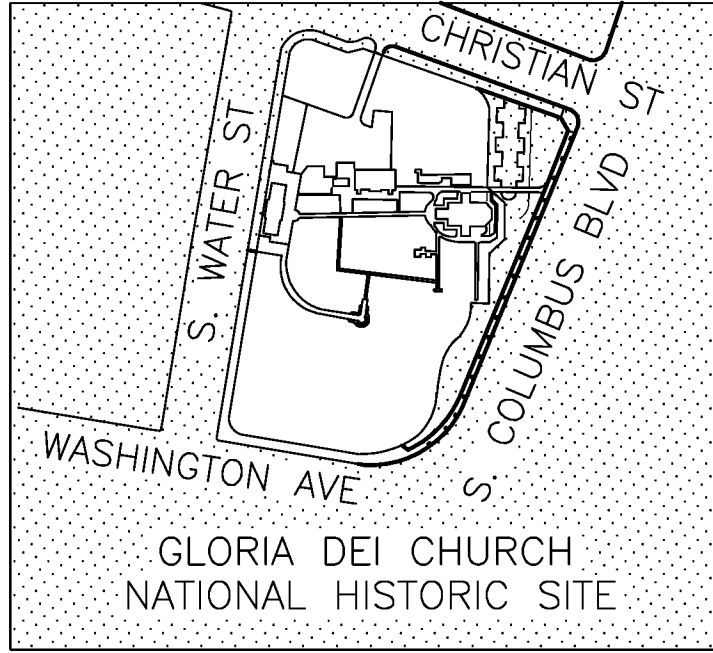
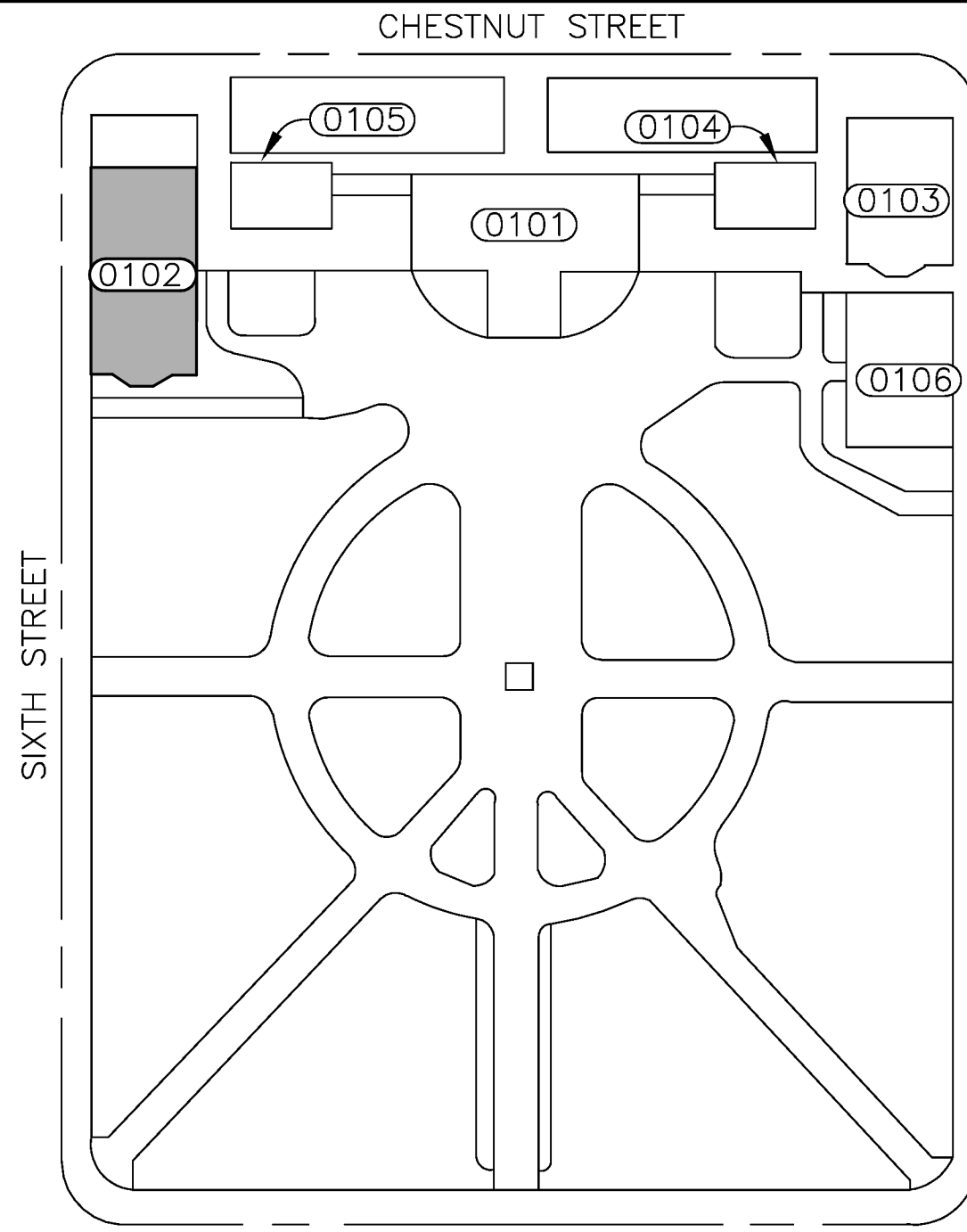
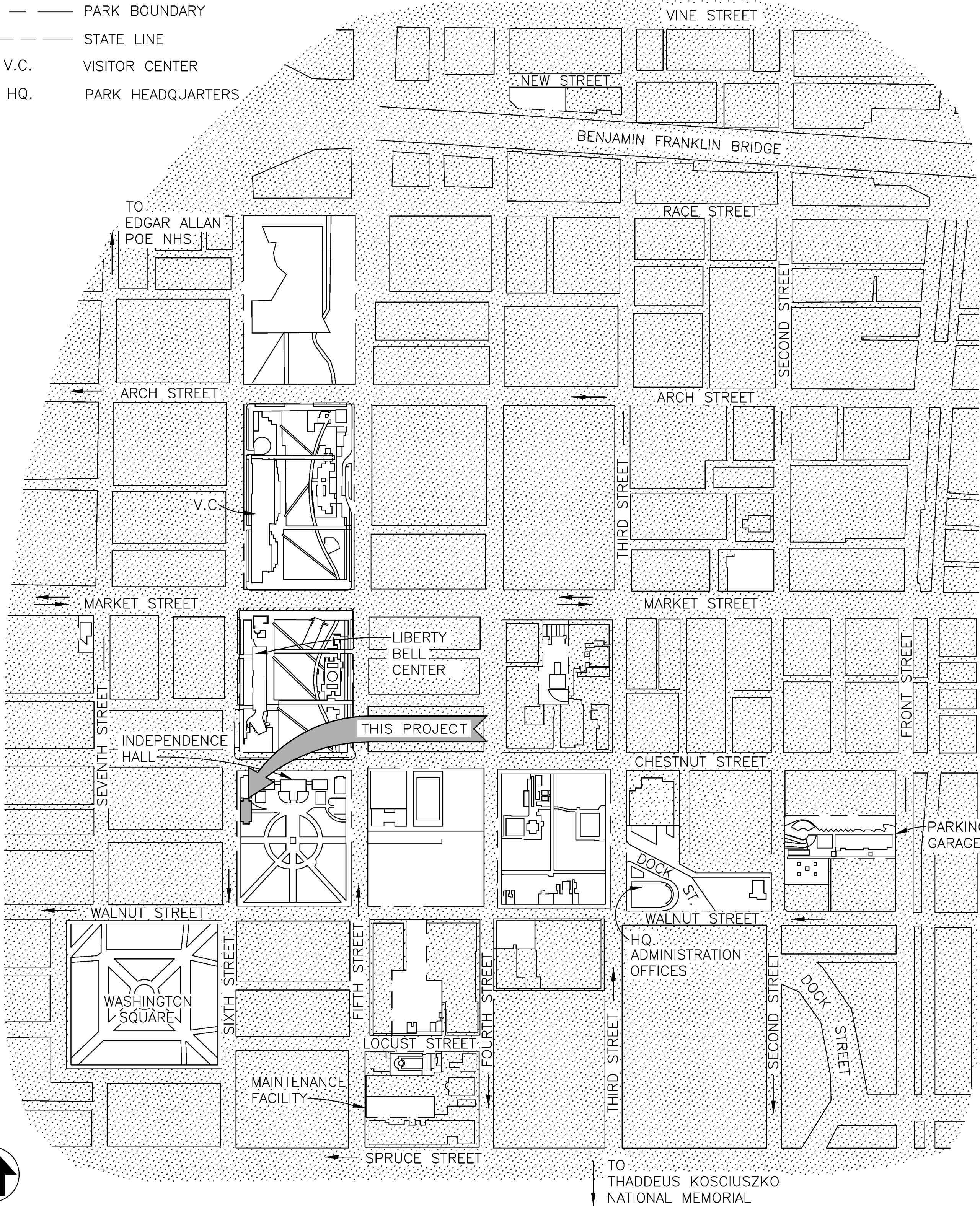


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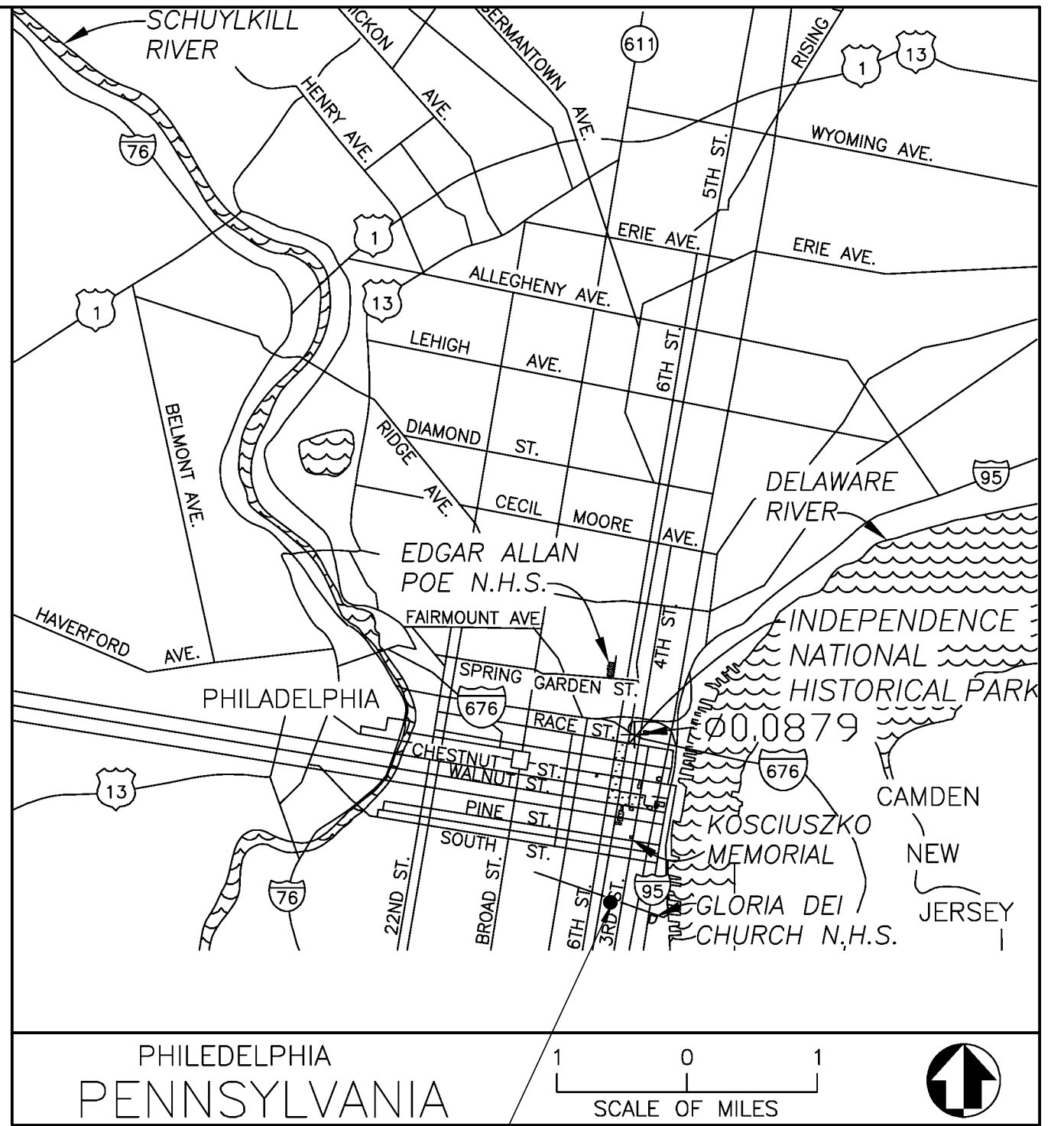
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- PARK BOUNDARY
- STATE LINE
- V.C. VISITOR CENTER
- HQ. PARK HEADQUARTERS

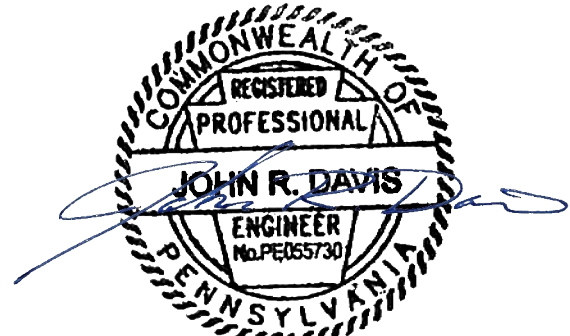
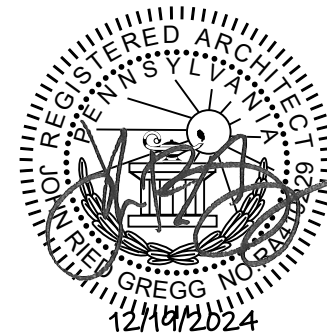


INDEPENDENCE SQUARE

- 0101 INDEPENDENCE HALL
- 0102 CONGRESS HALL
- 0103 OLD CITY HALL
- 0104 EAST WING
- 0105 WEST WING
- 0106 PHILOSOPHICAL HALL



SHEET	SUB SHEET	TITLE OF SHEET
1	G1.0	COVER SHEET
2	AS1.0	SITE PLAN
3	S0.0	GENERAL NOTES - STRUCTURE
4	S1.3	OVERALL ROOF KEY PLAN
5	S1.4	PARTIAL EAST ELEVATION
6	S2.1	CHIMNEY 1
7	S2.2	CHIMNEY 2
8	S2.3	CHIMNEY 3
9	S2.4	CHIMNEY 4
10	S2.5	CHIMNEY 5
11	S2.6	CHIMNEY 6
12	S3.0	CHIMNEY REINFORCEMENT
13	R3.1	WEST ELEVATION SELECT EXTERIOR RESTORATION
14	R3.2	EAST ELEVATION SELECT EXTERIOR RESTORATION
15	R3.3	ROOF PLAN SELECT EXTERIOR RESTORATION
16	R3.4	MASONRY RESTORATION DETAILS

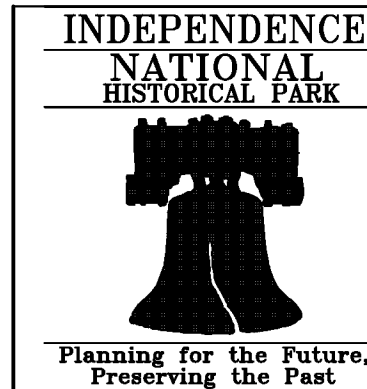


DECEMBER 19, 2024
100% FINAL CONSTRUCTION DOCUMENTS
PMIS 320146B



INDEPENDENCE NATIONAL HISTORICAL PARK
WORLD HERITAGE SITE

BASIC DATA: BOUNDARY MAP 391/92,000; PARK
MINI FOLDER 1989; REVISED & REDRAWN 1/95.
UPDATED 11/98. SGS.



A/E FIRM - 24013

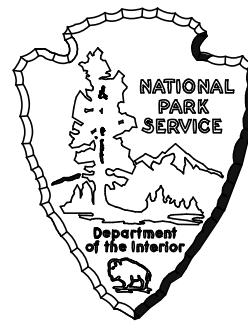
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GWWD, INC.
BALTIMORE, MD

STRUCTURAL:
KEAST & HOOD CO.
PHILADELPHIA, PA

CONSERVATOR:
BUILDING CONSERVATION
ASSOCIATES, INC.
PHILADELPHIA, PA

Mark	Sheet	REVISION	Date	Initial

QUALITY DESIGN CERTIFICATION	
☐	Prepared in Accordance with Design Development (Title I) Drawing No. _____
☐	OR Variance from Design Development (Title I) Approved by Superintendent on _____ Date _____
☐	OR Construction Drawing Not Preceded by Design Development (Title I) _____ Date _____
Project Manager	Date



100% FINAL
CONSTRUCTION DOCUMENTS

UNITED STATES
DEPARTMENT OF THE INTERIOR

DENVER SERVICE CENTER

TITLE OF DRAWING
REHABILITATE CONGRESS HALL MASONRY

LOCATION WITHIN PARK
CONGRESS HALL

NAME OF PARK
INDEPENDENCE NATIONAL HISTORICAL PARK

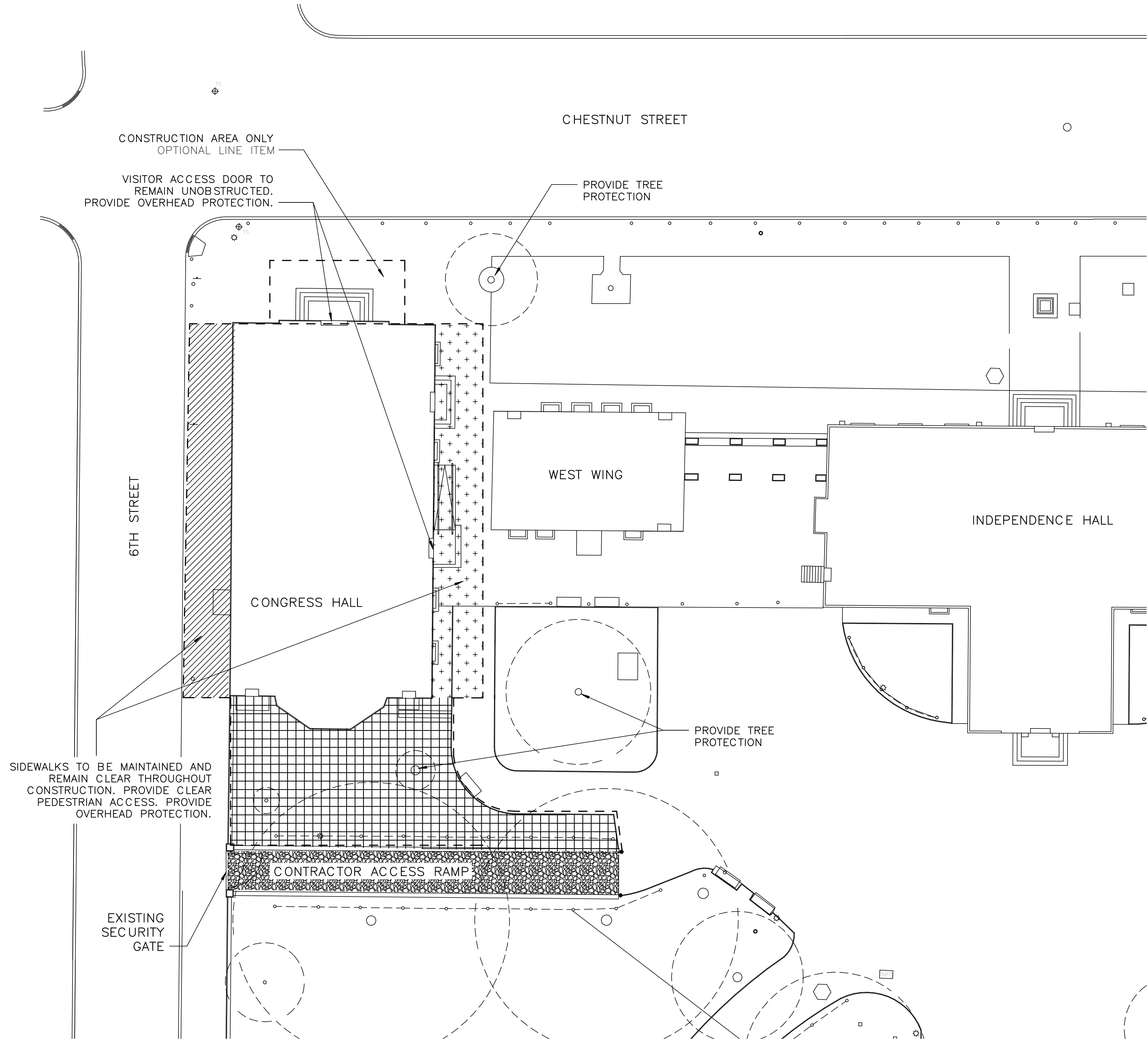
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NORTHEAST PHILADELPHIA PA

DRAWING NO.
391
190326

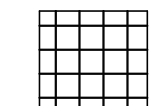
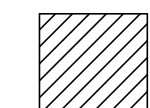
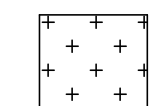
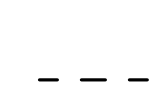
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320146B

SHEET
1 OF 15

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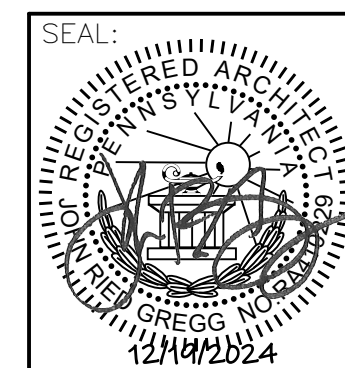
LEGEND

-  APPROX. STAGING AREA LOCATION
-  APPROX. AREA FOR LIFT USE
-  APPROX. AREA FOR SCAFFOLDING INSTALLATION
-  CONTRACTOR VEHICLE ACCESS (AVAILABLE OUTSIDE OF 9AM TO 5PM)
-  CONSTRUCTION LIMIT LINE

- Contractor shall meet the requirements of all codes, rules, and regulations governing this project as set by Federal, State, and local codes, building department agencies, and the Owner.
- Contractor shall meet the requirements of the applicable accessibility guidelines for all aspects of this project.
- The information included in these drawings with regards to the existing building is based on previous documentation provided to the Architect as well as what is readily observable at the project site. Destructive investigation techniques have not been used to verify information that is not readily observable.
- Contractor shall coordinate disconnection of mechanical, electrical, and plumbing systems before starting work on the chimneys and reconnection after the work is complete.

GENERAL NOTES

1 SITE PLAN
AS1.0 NO SCALE



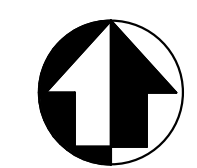
A/E FIRM -- 24013
PRIME/ARCH:
GWMO, INC.
BALTIMORE, MD

REVISIONS:			DESIGNED: IAG CADD: IAG TECH. REVIEW: JRG DATE: 12/19/24
NO.	DATE	DESCRIPTION	

SUB SHEET NO.
AS1.0

TITLE OF SHEET
SITE PLAN

REHABILITATE CONGRESS HALL MASONRY
INDEPENDENCE NATIONAL HISTORICAL PARK



DRAWING NO.
391
190326
PMIS/PKG NO.
320146B
SHEET
2 OF **15**

GENERAL

- These notes highlight but do not replace the specifications contained in the Project Manual.
- The applicable building code is 2024 IEBC, 2024 IBC as amended by the Denver Service Center Design Standards.
- Refer to the Design Loads and Factors table, this sheet for code required loads for the project.
- Contractor shall be responsible for the proper and safe design of shoring systems for trenches and excavations.
- Notify Contracting Officer in case of discrepancies between drawings, and these notes, and Project Manual before proceeding with the work.
- If conditions disclosed during demolition reveal unforeseen conditions, promptly request direction from Contracting Officer before proceeding.
- Do not scale drawings.
- Sections and details shown, while drawn for specific locations, are intended to establish the general types of details to be used throughout.
- The SEOR's review of a submittal shall not relieve the Contractor of their responsibility to follow the intent of the contract drawings.

DELEGATED DESIGN ITEMS

- Employ or retain a licensed professional engineer in the project jurisdiction to design and detail the following performance specified structural components:
 - Shoring / Scaffolding
- See individual material sections for additional requirements.

EXISTING BUILDING/STRUCTURE NOTES

- Verify all existing dimensions and elevations before beginning work.
- Utmost care shall be exercised at all times when working on existing structural members and masonry bearing walls to avoid impairing the carrying capacity of the existing structure.
 - Should the Contracting Officer determine that the carrying capacity of the existing structure has been impaired by, or as a result of, the operations of the Contractor, or is otherwise not in conformance with the contract documents, appropriate remedial work shall be required.
 - Any damage resulting from the operations of the Contractor shall be repaired as directed by the Contracting Officer at no additional cost to the Contracting Officer.
- Notify the Contracting Officer before cutting or removal of any part of the existing structure not indicated to be modified or demolished.
- Burning or welding in the building is strictly prohibited.
- Definitions:
 - Remove: Detach items from existing construction and dispose of them off-site unless indicated to be salvaged or reinstalled.
 - Remove and Salvage: Detach items from existing construction, in a manner to prevent damage, and store.
 - Remove and Reinstall: Detach items from existing construction, in a manner to prevent damage, prepare for reuse, and reinstall where indicated.
 - Remove and Replace: Detach items from existing construction and replace in kind.
 - Existing to Remain: Leave existing items that are not to be removed and that are not otherwise indicated to be salvaged or reinstalled.
 - Dismantle: To remove by disassembling or detaching an item from a surface, using gentle methods and equipment to prevent damage to the item and surfaces; disposing of items unless indicated to be salvaged or reinstalled.
- Work shall be completed in a manner consistent with the Secretary of Interior's Standard for Preservation.
- The Contractor is responsible to provide protection of open chimneys after caps are removed and until they are replaced from water, dust, and debris.

STRUCTURAL STEEL

- Detail, fabricate, and erect structural steel in accordance with AISC 360 and AISC 303 "Code of Standard Practice".
- The structural steel detailer shall develop all elevations, plans, and sections without reproducing same from the design drawings. Detailer shall adequately cross reference the structural drawings to the satisfaction of the SEOR. All shop drawings shall be submitted for review and review comments incorporated prior to fabrication.
- Materials shall conform to the following:
 - Hollow Structural Sections (HSS) - ASTM A500, Gr C (Fy = 50 ksi)
 - Plates 4" thickness or less - ASTM A572, Gr 50 (Fy = 50 ksi)
 - Welding Electrodes - AWS A5.1 E70XX
- Field cutting or burning of structural steel is prohibited.
- Galvanize all steel in accordance with ASTM A-123. Repair scratched or abraded galvanized surfaces with zinc-rich coating. After galvanizing, straighten members to meet AISC standard mill tolerances.

CAST-IN-PLACE LINER

- All installers of chimney lining system shall be properly trained, equipped, and certified by manufacturer.
- Cast-in-place cement lining material to be tested/listed to the UL 1777 standards.
- Provide light-weight cementitious lining material with aggregates conforming to ASTM C332 with Fc = 2000 psi per ASTM C495.
- Submit product data sheets for all components of system prior to use.

SHORING/SCAFFOLDING

- The contractor shall submit shoring/scaffolding shop drawings and calculations, prepared and sealed by a professional engineer licensed in the state of the project, for the following:
 - Building/structure support shoring.
 - Scaffolding.
- Shoring and scaffolding shall comply with O.S.H.A. regulations.
- The SEOR will review shoring/scaffolding submittals only for loads transmitted to the building structure. Submittals shall clearly indicate the location and magnitude of all loads applied to the building or structure. The contractor is responsible for design and performance of the shoring/scaffold system.

STANDARD ABBREVIATIONS

ABV	Above	DIAG	Diagonal	JT	Joint	RD	Roof Drain
ADDL	Additional	DIM(S)	Dimension(s)	k	Kip (thousand pounds)	RDK	Roof Deck
ADH	Adhesive	DIR	Direction	KB	Knee Brace	REF	Reference
ADJ	Adjacent, Adjustment	DK	Deck	LB, #	Pound	REINF	Reinforce(d)/ Reinforcement
AESS	Architecturally Exposed Structural Steel	DL	Dead Load	LDGR	Ledger (board)	REQ(D)	Require(d)
		DN	Down	LEN	Length	RET	Retaining
AFF	Above Finished Floor	do.	Ditto	LG	Long	REV	Revise(d)/Revision
AGG	Aggregate	DP	Deep	LI	Linear Inch	RO	Rough Opening
ALT	Alternate	DS	Drilled Shaft,	LL	Live Load	RQMT(S)	Requirement(s)
ANCH	Anchor		Drag Strut	LLBB	Long Legs Back to Back	RXN	Reaction
APRX	Approximate	DWG(S)	Drawing(s)	LLH	Long-Leg Horizontal	SC	Slip-Critical (connection)
AR	Anchor Rod	DWL	Dowel	LLV	Long-Leg Vertical	SCHED	Schedule
ARCH	Architect(ural)	(E)	Existing	LOCN	Location	SECT	Section
BOT, B	Bottom	EA	Each	LONGIT	Longitudinal	SEOR	Structural Engineer of Record
B/	Bottom of	EE	Each End	LP	Low Point	SF	Square Foot
B-B	Back-to-Back	EF	Each Face	LSH	Long Side Horizontal	SFRM	Spray-Applied Fire Resistant Material
BAL	Balance	EJ	Expansion Joint	LSL	Long-Slotted Hole,	SHRG	Shoring
BCE	Bot Chord Extension	EL	Elevation	LSV	Long Side Vertical	SHTHG	Sheathing
BEL	Below	ELEC	Electrical	LVL	Laminated Veneer Lumber	SIM	Similar
BF	Braced Frame	ELEV	Elevator	LW	Long Way	SL	Slope(d)
BLDG	Building	EMBED	Embedment/Embedded	LWC	Light Weight Concrete	SLBB	Short Legs Back to Back
BLK(G)	Block(ing)	ENGR	Engineer	M	Moment	SLV	Sleeve
BM	Beam	EOD	Edge Of Deck	MAS	Masonry	SOD	Slab On Deck
BP	Base Plate	EOS	Edge Of Slab	MATL	Material	SOG	Slab On Grade
BR	Brace(ing)	EQ	Equal	MAX	Maximum	SPA	Space(s)/Spacing
BRDG	Bridging	EQP	Equipment	MECH	Mechanical	SPEC(S)	Specification(s)
BRG	Bearing	ES	Each Side	MFR	Manufacturer	SQ	Square
BRK	Brick	EXCAV	Excavate/Excavation	MIN	Minimum	SS	Stainless Steel
BRKT	Bracket	EXP	Expansion	MISC	Miscellaneous	SSD	Saturated Surface-Dry
BS	Both Sides	EXT	Exterior	MO	Masonry Opening	SSL	Short-Slotted Hole
BSMT	Basement	EW	Each Way	MONO	Monolithic	STD	Standard
BT	Bent	F/	Face of	MTL	Metal	STIFF	Stiffener
BTWN	Between	FAB	Fabricate	NIC	Not In Contract	STIR	Stirrup
BYD	Beyond	FDN	Foundation	NO.	Number	STL	Steel
CANT	Can'tiever	FIN.	Finish(ed)	NOM	Nominal	STRUC	Structural
CAP.	Capacity	FL	Floor	NS	Near Side	SUPT	Support
C-C	Center-to-Center	FLG	Flange	NTS	Not To Scale	SW	Short Way
CE	Concrete Encased	FP	Fireproofing	NWC	Normal Weight Concrete	SYM	Symmetrical
CFMF	Cold-Formed Metal Framing	FRMG	Framing	OC	On Centers	T	Top
CHAM	Chamfer	FRP	Fiber-Reinforced Polymer	OD	Outside Diameter	T/	Top of
CI	Cast Iron	FS	Far Side	O.F.	Outside Face	T&B	Top and Bottom
CIP	Cast In Place	FT	Feet	O-O	Out-to-Out	TC	Terra Cotta
CJ	Control/Contraction Joint	FTG	Footing	OPG	Opening	TCE	Top Chord Extension
CL	Centerline	FUT	Future	OPP	Opposite	TEMP	Temporary, Temperature
CLG	Ceiling	GA	Gage	OSB	Oriented Strand Board	T&G	Tongue and Groove
CLIN	Contract Line Item Number	GALV	Galvanize(d)	OSL	Outstanding Leg,	THD	Thread(ed)
CLR	Clear	GC	General Contractor	OVS	Oriented Strand Lumber	THK	Thick(ness)
CM	Construction Manager	GR	Grade	PAF	Oversized-Hole	TLS	Tension Lap Splice
CMU	Concrete Masonry Unit	GYP	Gypsum	PC	Powder Actuated Fastener	TRANS	Transverse
COL	Column	HDR	Header	PCF	Precast Concrete, Piece	TYP	Typical
COMB.	Combined	HEF	Horizontal Each Face	PCF	Pounds per Cubic Foot	UNO	Unless Noted Otherwise
COMP	Composite	HGR	Hanger	PED	Pedestal	U-P	Underpinning
CONC	Concrete	HI	High	PERP	Perpendicular	V	Shear
COND	Condition	HIF	Horizontal Inside Face	PREFAB	Prefabricate(d)	VAR	Varies
CONN	Connection	HK	Hook(ed)	PL	Plate	VB	Vapor Barrier
CONST	Construction	HOF	Horizontal Outside Face	PLF	Pounds per Lineal Foot	VEF	Vertical Each Face
CONT	Continuous	HORIZ	Horizontal	PLYWD	Plywood	VERT	Vertical
CONTR	Contractor	HP	High Point	PMF	Premolded Filler	VIF	Vertical Inside Face,
COORD	Coordinate	HSS	Hollow Strcutral Steel	PROJ	Project(ion)	VOF	Vertical Outside Face
COV	Cover	HT	Height	PROV	Provide	VR	Vapor Retarder
CP	Cap Plate	H&V	Horizontal & Vertical	PSF	Pounds per Square Foot	w/	With
CSIA	Chimney Safety Inst. of America	HVAC	Heating, Ventilating, & Air Conditioning	PSI	Pounds per Square Inch	w/o	Without
CSK	Countersunk	I.D.	Inside Diameter	PSL	Parallel Strand Lumber	WD	Wood
CTR	Center(ed)	I.F.	Inside Face	PT	Preservative-Treated,	WI	Wrought Iron
CY	Cubic Yard	INFO	Information	PVC	Polyvinyl Chloride	WP	Work Point, Waterproofing
DBL	Double	INSTL	Instal/Installation	QTY	Quantity	WT	Weight
DEMO, (D)	Demolition/Demolish	INSUL	Insulation	R	Radius	WWF	Welded Wire Fabric
DEPR	Depress(ed)/Depression	INT	Interior	RCMU	Reinforced Concrete Masonry Unit	XS	Extra Strong
DET(S)	Detail(s)	JST(S)	Joist(s)			XXS	Double Extra Strong
DEV	Develop/Development					±	Exist Dim or El (VIF)
DIA, Ø	Diameter						

STRUCTURAL DESIGN LOADS AND FACTORS

DESIGN CODE: IBC 2024 ED
ASCE 7-22

WIND LOAD DATA	
RISK CATEGORY	II
WIND EXPOSURE	C
ULTIMATE WIND SPEED (Vult) (MPH)	114
NOMINAL WIND SPEED (Vasd) (MPH)	90
INTERNAL PRESSURE COEFFICIENT	± 0.18
DESIGN WIND PRESSURE ON CHIMNEY	+44psf/-59psf

NOTES:

- NO CHANGES TO LIVE OR SUPERIMPOSED DEAD LOAD FOR STRUCTURE.
- WIND LOADING CONTROLS LATERAL LOAD ON AREAS OF SCOPE.

MATERIAL LEGEND

DEMO (D)	EXISTING (E)	NEW	
			FOUNDATION
			UNDERPINNING
			CONCRETE
			BRICK
			STONE
			CMU
			BEARING WALL (ANY MATL) BEL
			CONC BM BELOW
			STEEL BEAM/ COL
			JOIST

CONSTRUCTION NOTATION

WORK INDICATED IS NEW UNLESS NOTED AS
(E) = EXISTING CONSTRUCTION TO REMAIN.
(D) = DEMOLISH EXISTING CONSTRUCTION.
(R) = REPLACE EXISTING CONSTRUCTION WITH ____.

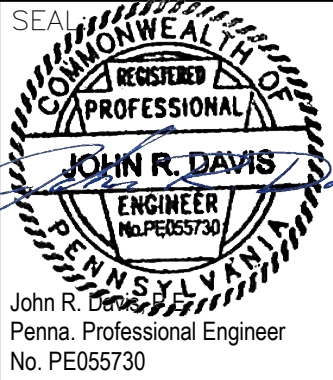
SYMBOLS

	SLOPE ARROW
	APPROXIMATE TEMPORARY SHORE LOCATION
	DIMENSION. DOT SNAPS TO CL OR GRIDS. TICK SNAPS TO EDGE OR FACE OF OBJECT.
	(E) GRID LINE

STRUCTURAL DRAWINGS LIST

SHEET NUMBER	DRAWING TITLE	100% Design Development - 2024-09-06	100% Construction Documents - 2024-11-08	100% Final Construction Documents - 2024-12-19
S0.0	GENERAL NOTES - STRUCTURE	•	•	•
S1.3	OVERALL ROOF KEY PLAN	•	•	•
S1.4	PARTIAL EAST ELEVATION		•	•
S2.1	CHIMNEY 1	•	•	•
S2.2	CHIMNEY 2	•	•	•
S2.3	CHIMNEY 3	•	•	•
S2.4	CHIMNEY 4	•	•	•
S2.5	CHIMNEY 5	•	•	•
S2.6	CHIMNEY 6	•	•	•
S3.0	CHIMNEY REINFORCEMENT	•	•	•

NO SCALE



A/E FIRM - 24013
PRIME/ARCH:
GWWO, INC.,
BALTIMORE, MD
STRUCTURAL ENG:
KEAST & HOOD,
PHILADELPHIA, PA

REVISIONS:		
NO.	DATE	DESCRIPTION

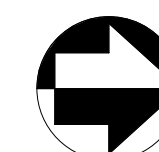
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TECH. REVIEW: JRD
DATE: 12/19/2024

SUB SHEET NO.
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TITLE OF SHEET
GENERAL NOTES –
STRUCTURE

REHABILITATE CONGRESS HALL MASONRY
INDEPENDENCE NATIONAL HISTORICAL PARK

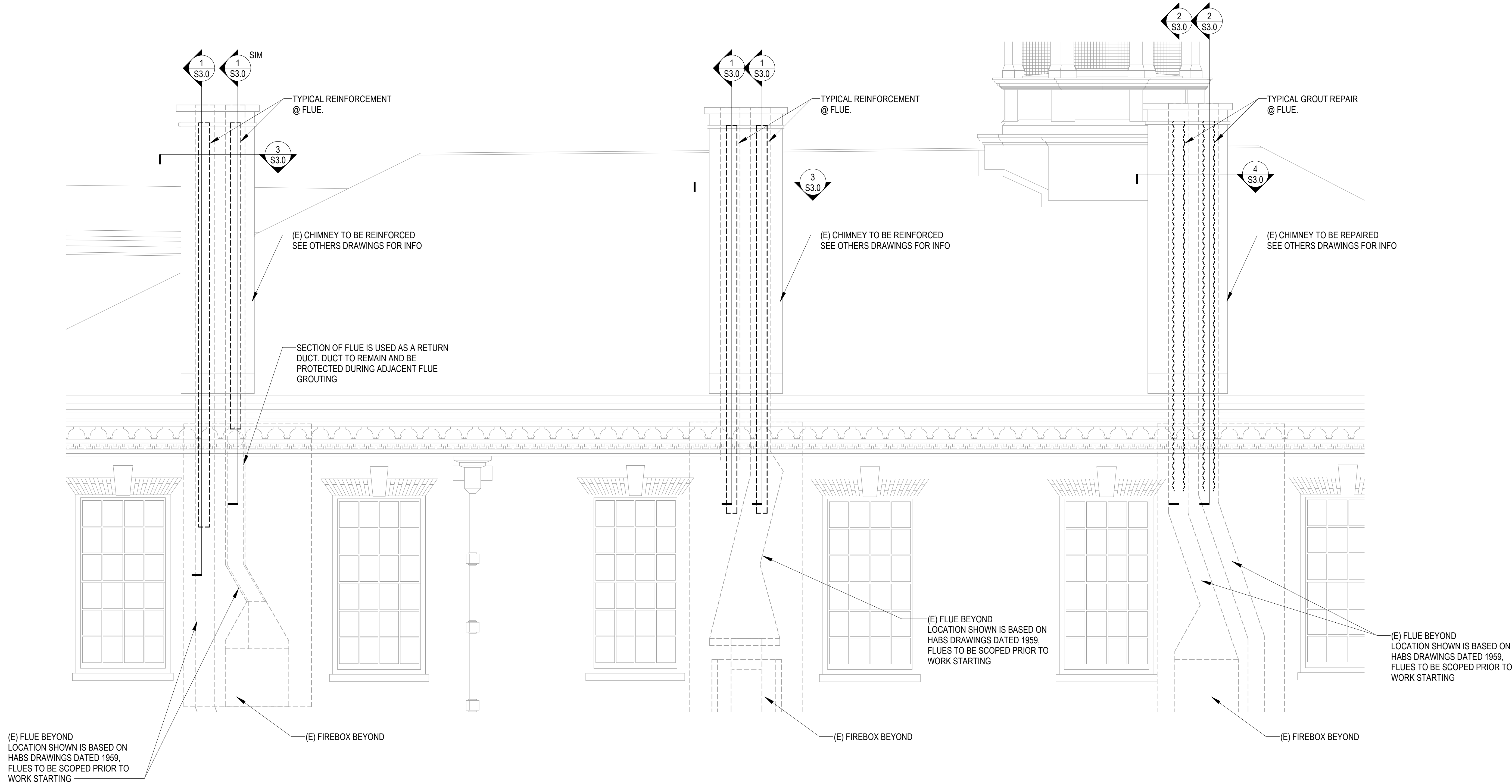
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190326
PMIS/PKG NO. 320146B
SHEET 3 OF 15



8 0 8 1
SCALE OF FEET

 John R. Davis Penna. Professional Engineer No. PE055730	A/E FIRM -- 24013	PRIME/ARCH: GWWD, INC. BALTIMORE, MD	STRUCTURAL ENG: KEAST & HOOD PHILADELPHIA, PA	REVISIONS:			DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
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							CADD:			
							Author			
							TECH. REVIEW:			
				Checker	S1.3	REHABILITATE CONGRESS HALL MASONRY INDEPENDENCE NATIONAL HISTORICAL PARK	391 190326			
				DATE:				PMIS/PKG NO.		
				12/19/2024					320146B	
						SHEET	4 OF 15			

12/16/2024 2:42:07 PM C:\Users\ckronenwetter\Documents\10053B Congress Hall Chimney R24_ckronenwetterAFZKK.rvt



1 ELEVATION @ CHIMNEY 1, CHIMNEY 2, & CHIMNEY 3
S1.4 SCALE (A)

SCALE (A) 2 0 2 4 6
SCALE OF FEET

SEAL
JOHN R. DAVIS
REGISTERED PROFESSIONAL ENGINEER
No. PE055730
Penna. Professional Engineer
No. PE055730

A/E FIRM - 24013
PRIME/ARCH:
GWOW, INC.
BALTIMORE, MD
STRUCTURAL ENG:
KEAST & HOOD
PHILADELPHIA, PA

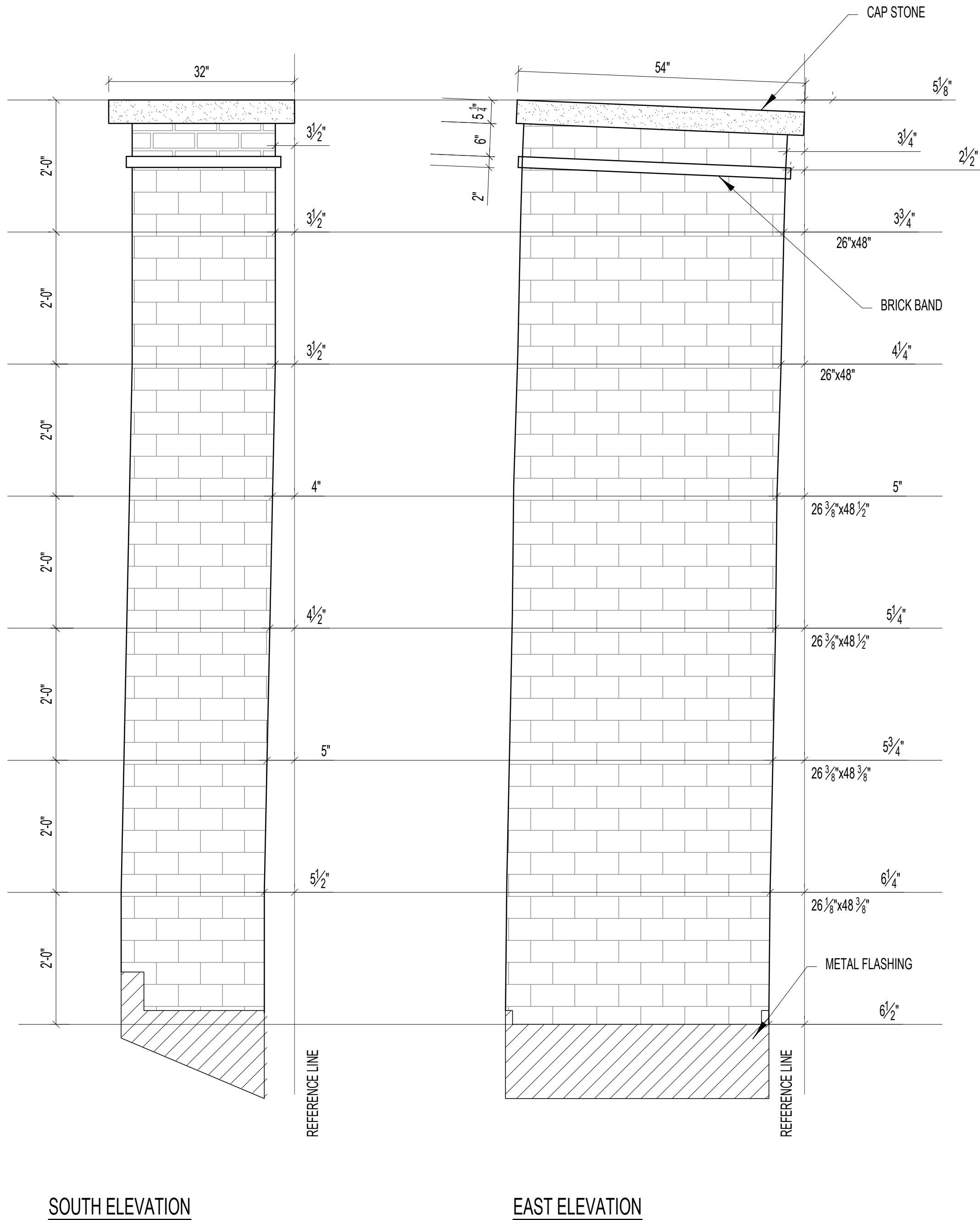
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DESIGNED:
Designer
CADD:
Author
TECH. REVIEW:
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12/19/2024

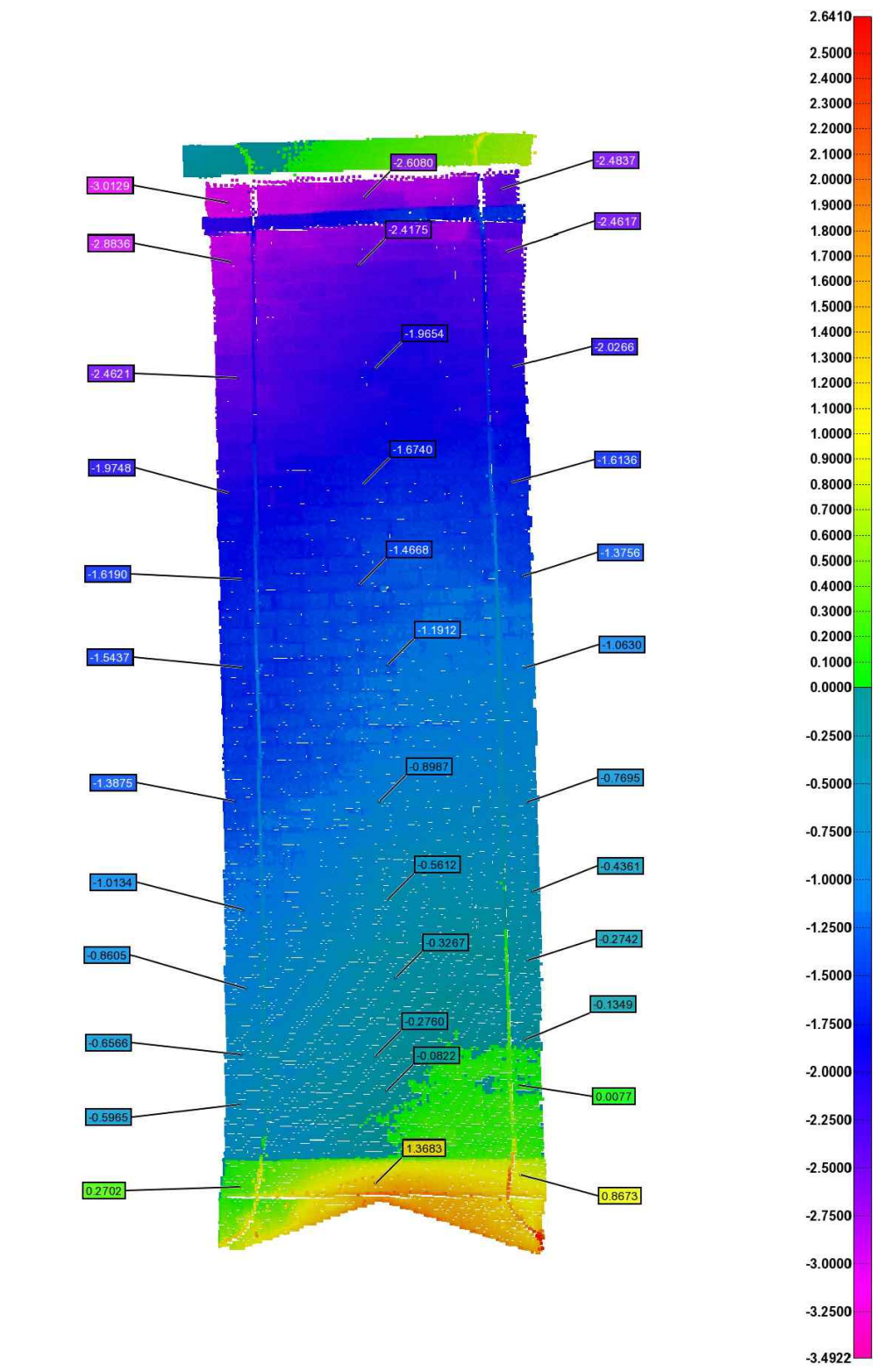
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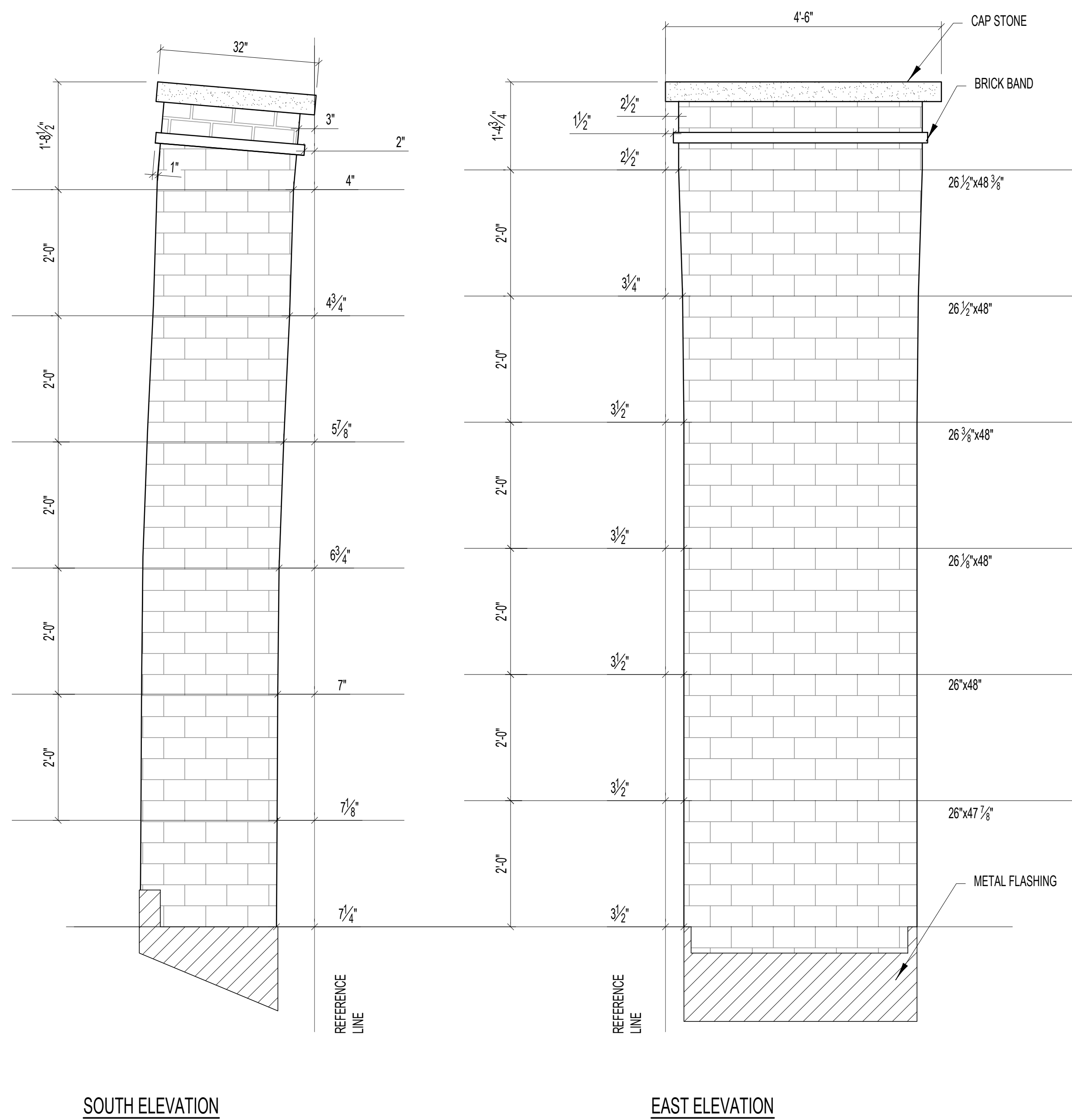
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PARTIAL EAST ELEVATION
REHABILITATE CONGRESS HALL MASONRY
INDEPENDENCE NATIONAL HISTORICAL PARK

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PMIS/PKG NO.
320146B
SHEET
5 OF 15

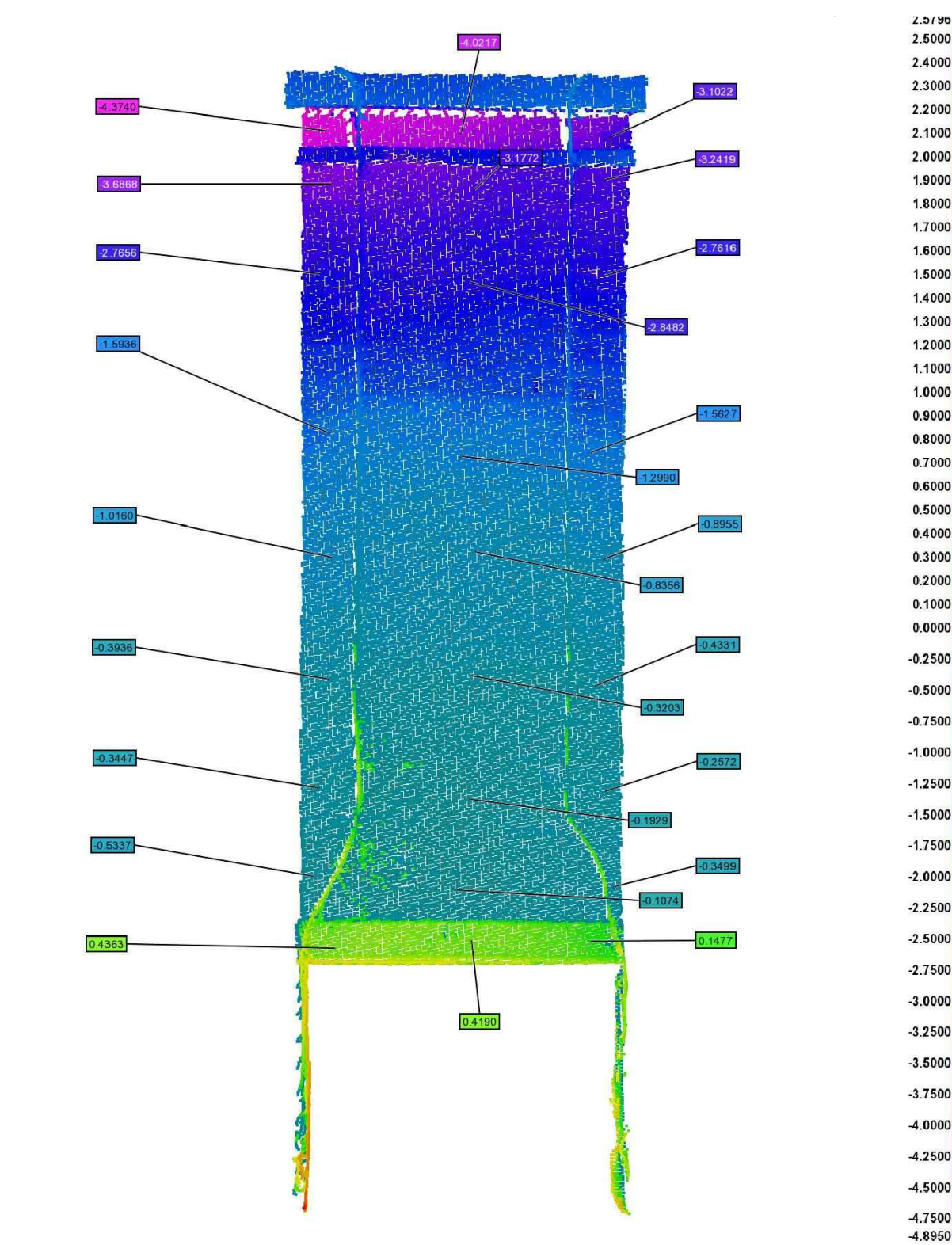


1 K&H SURVEY MEASUREMENTS
S2.1 NTS

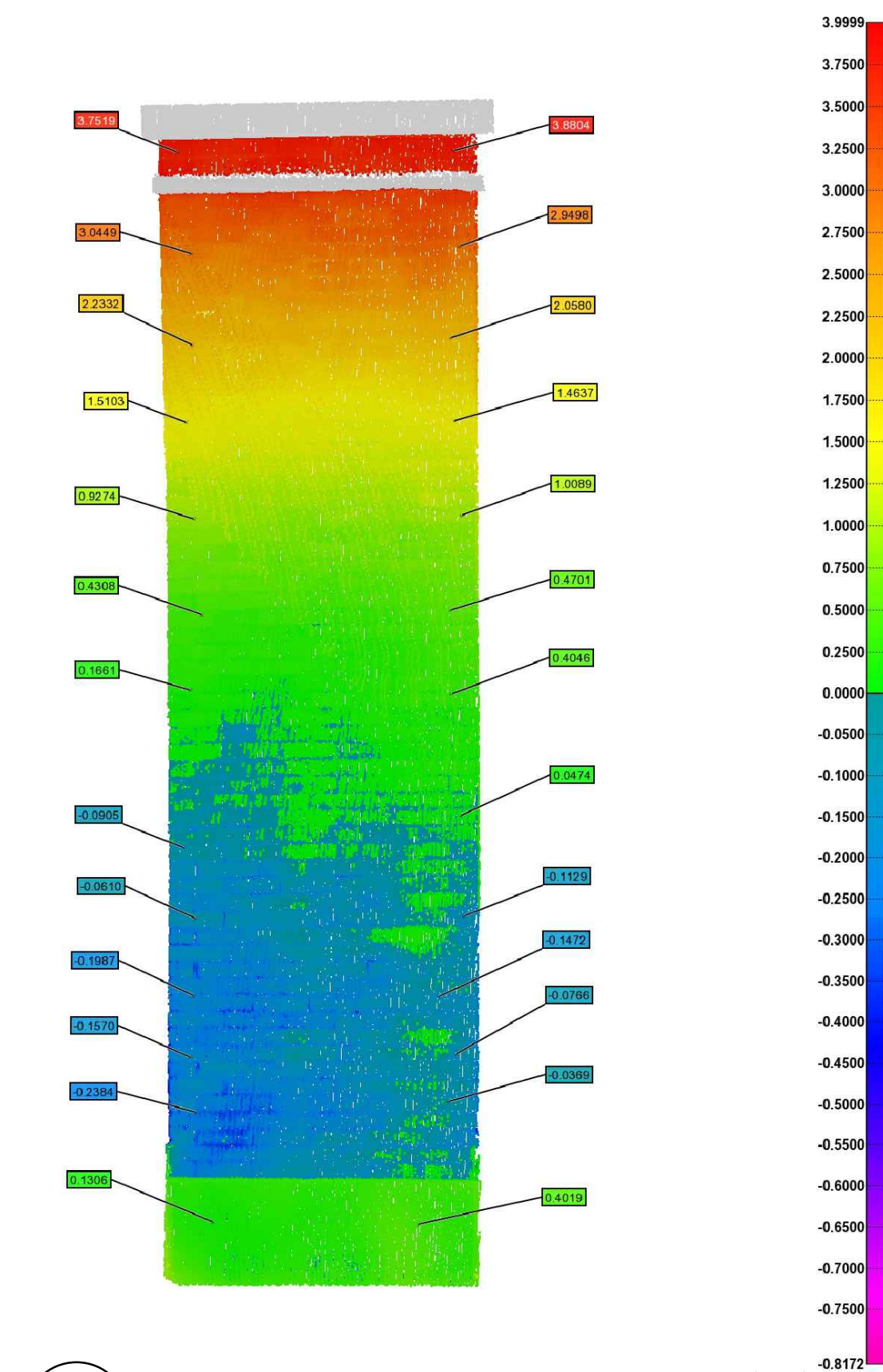




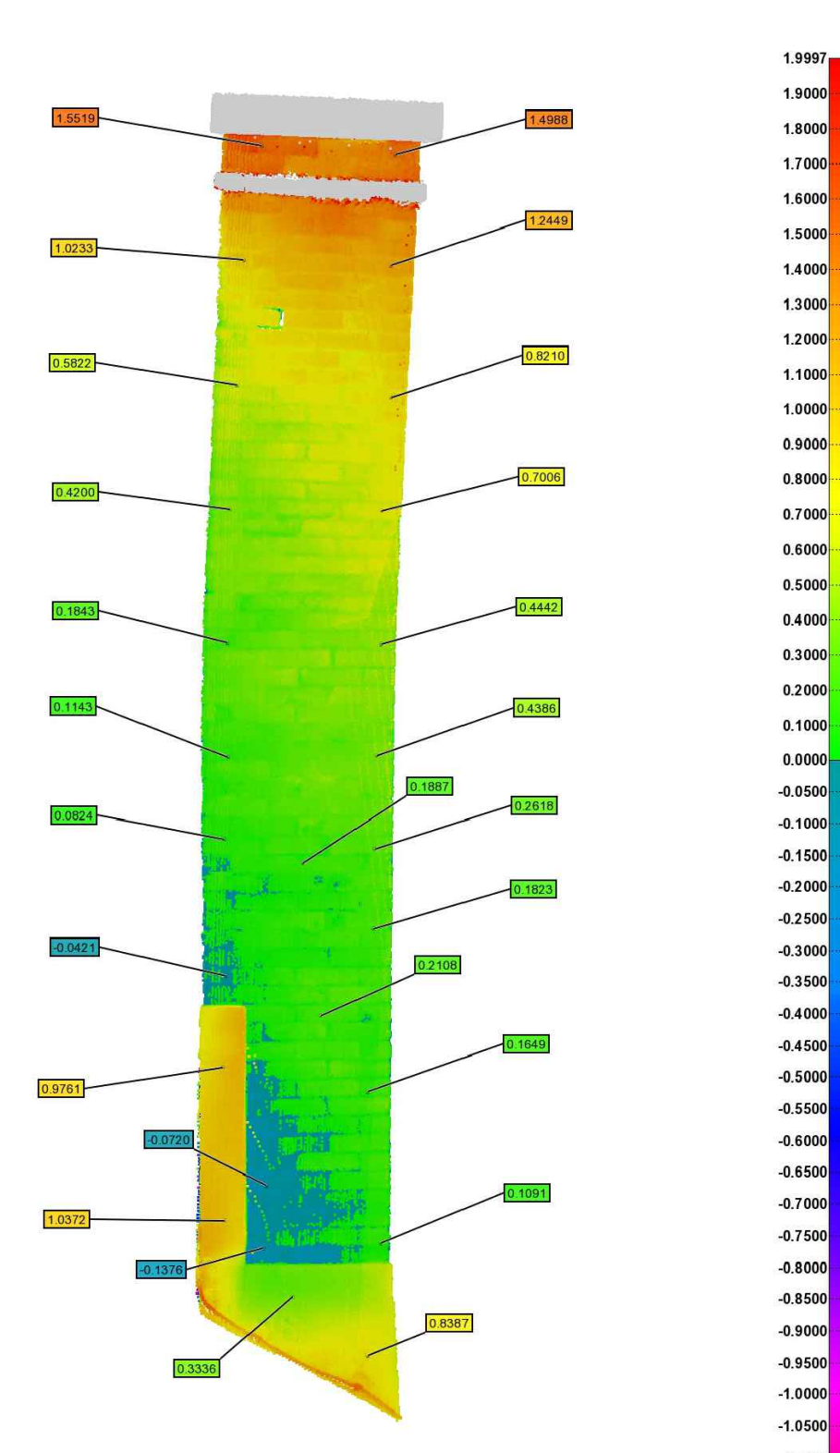
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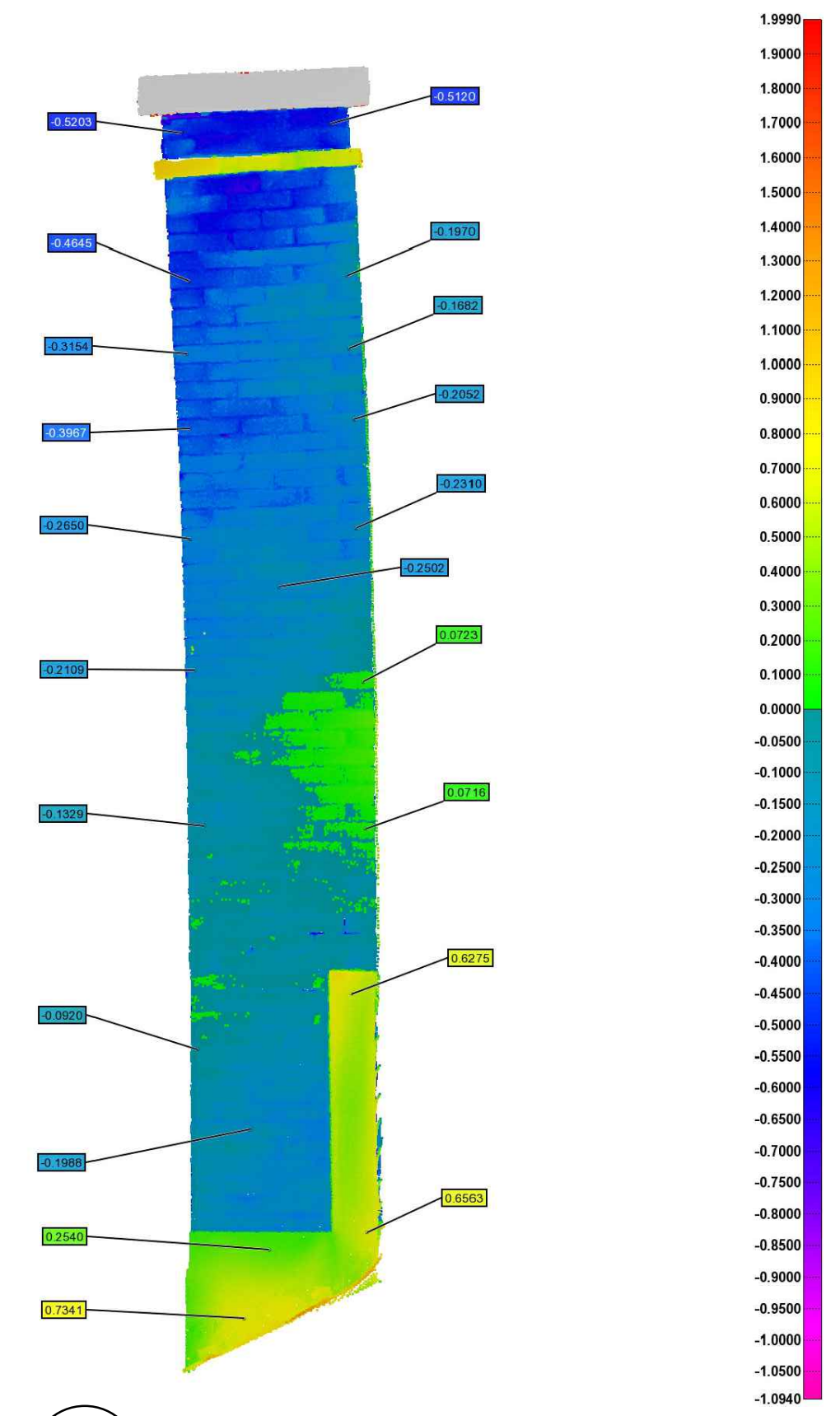
2 HEAT MAP WEST FACE
S2.2 NTS



4 HEAT MAP EAST FACE
S2.2 NTS



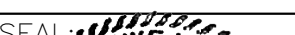
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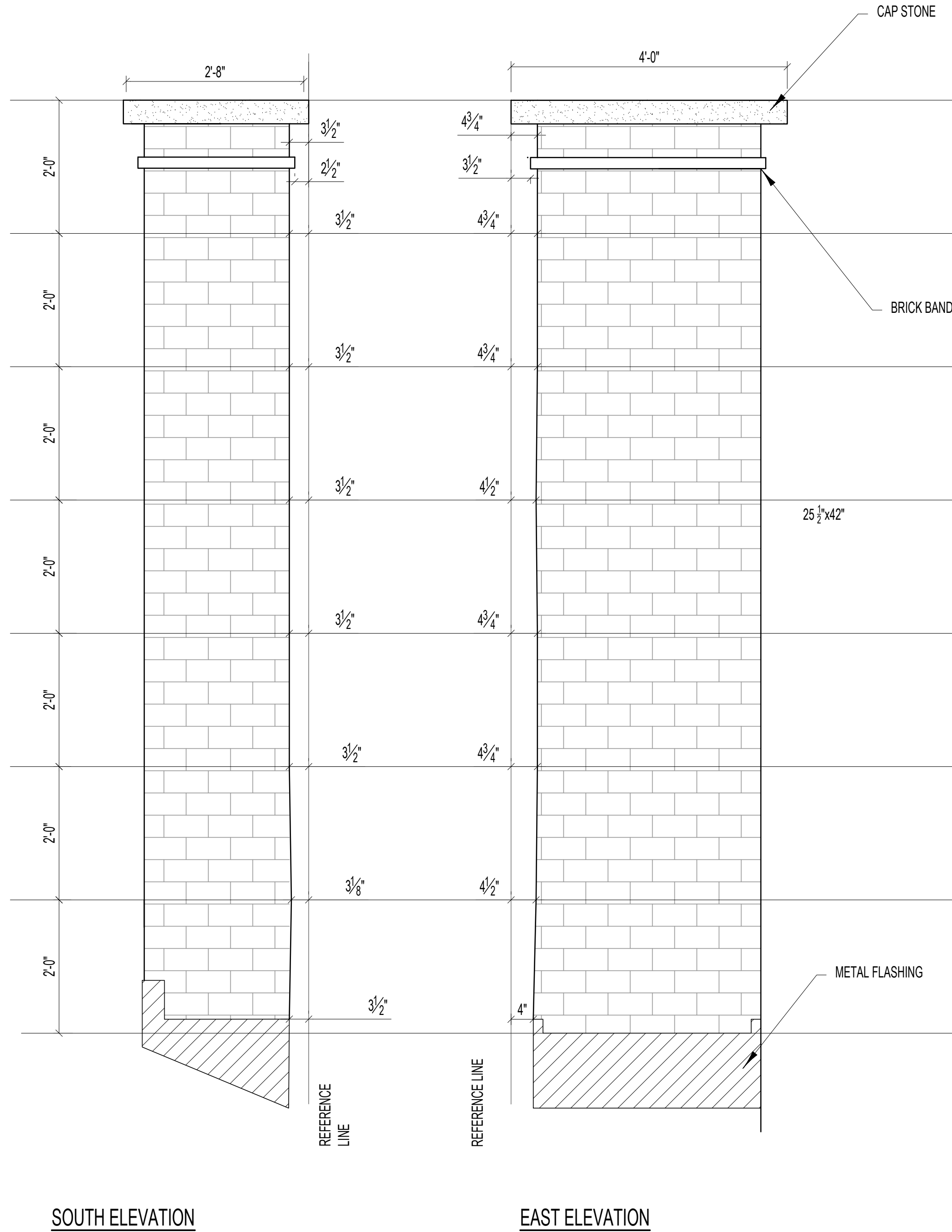


5 HEAT MAP NORTH FACE
S2.2 NTS

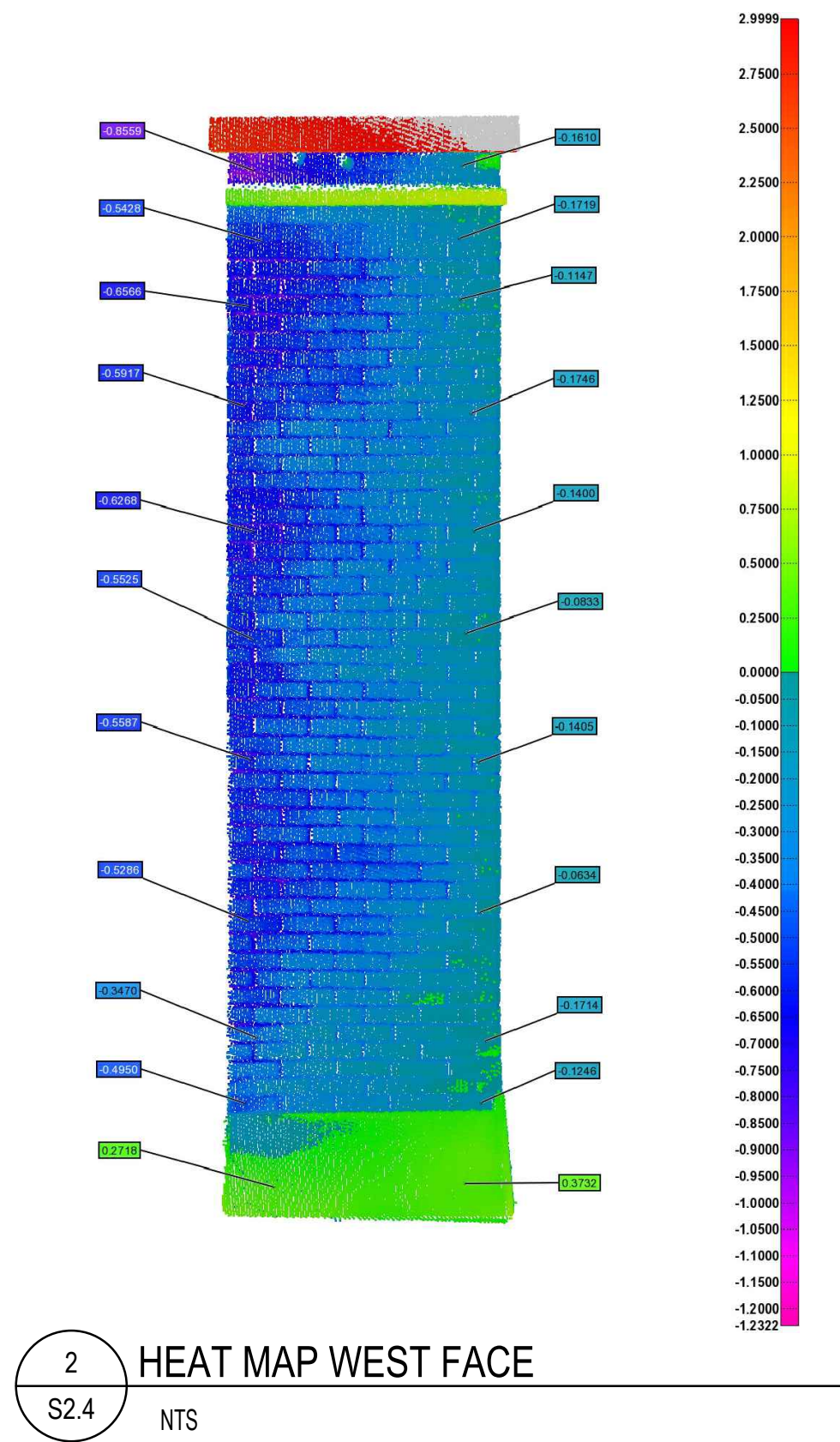
HEAT MAP LEGEND
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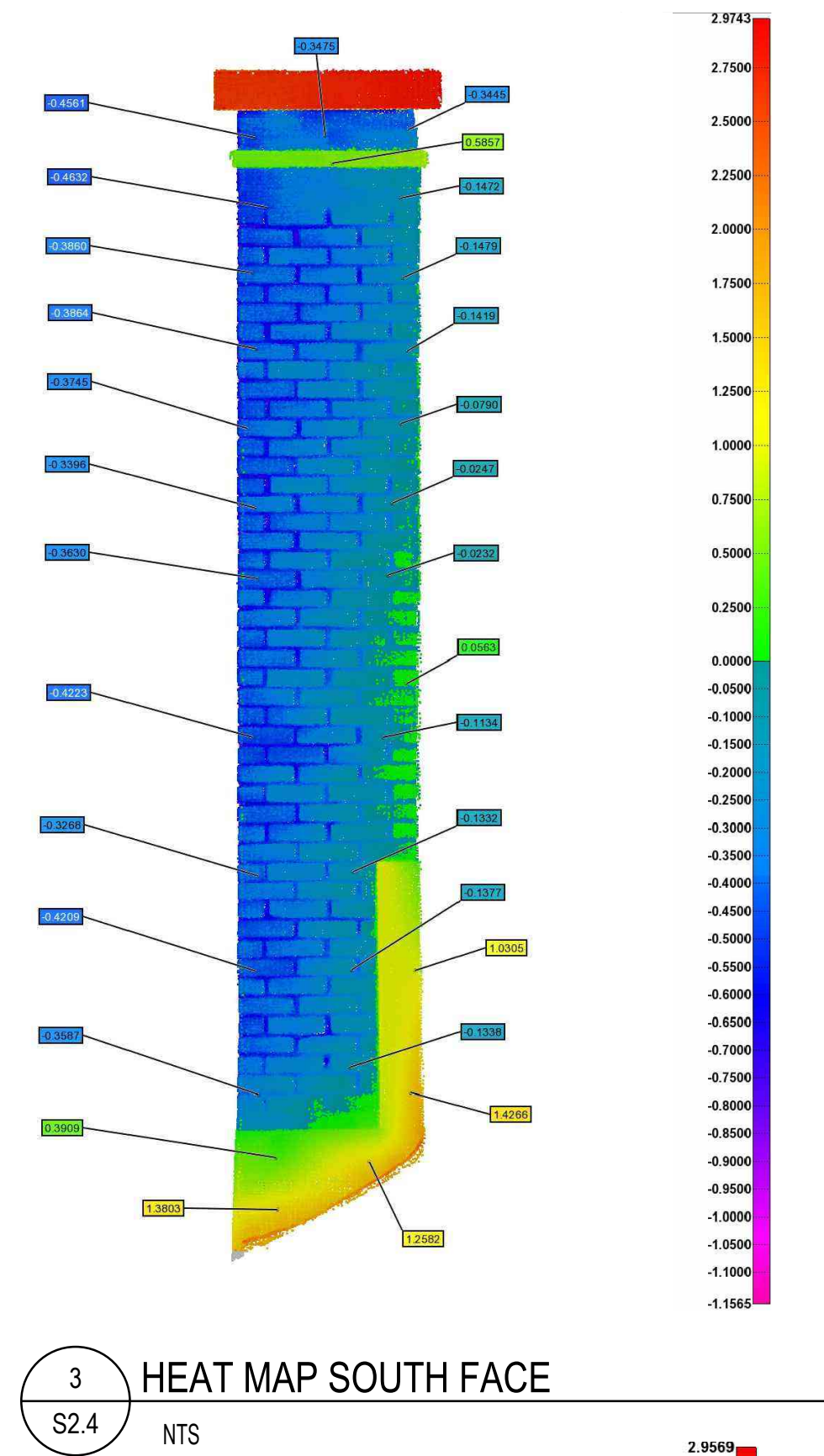
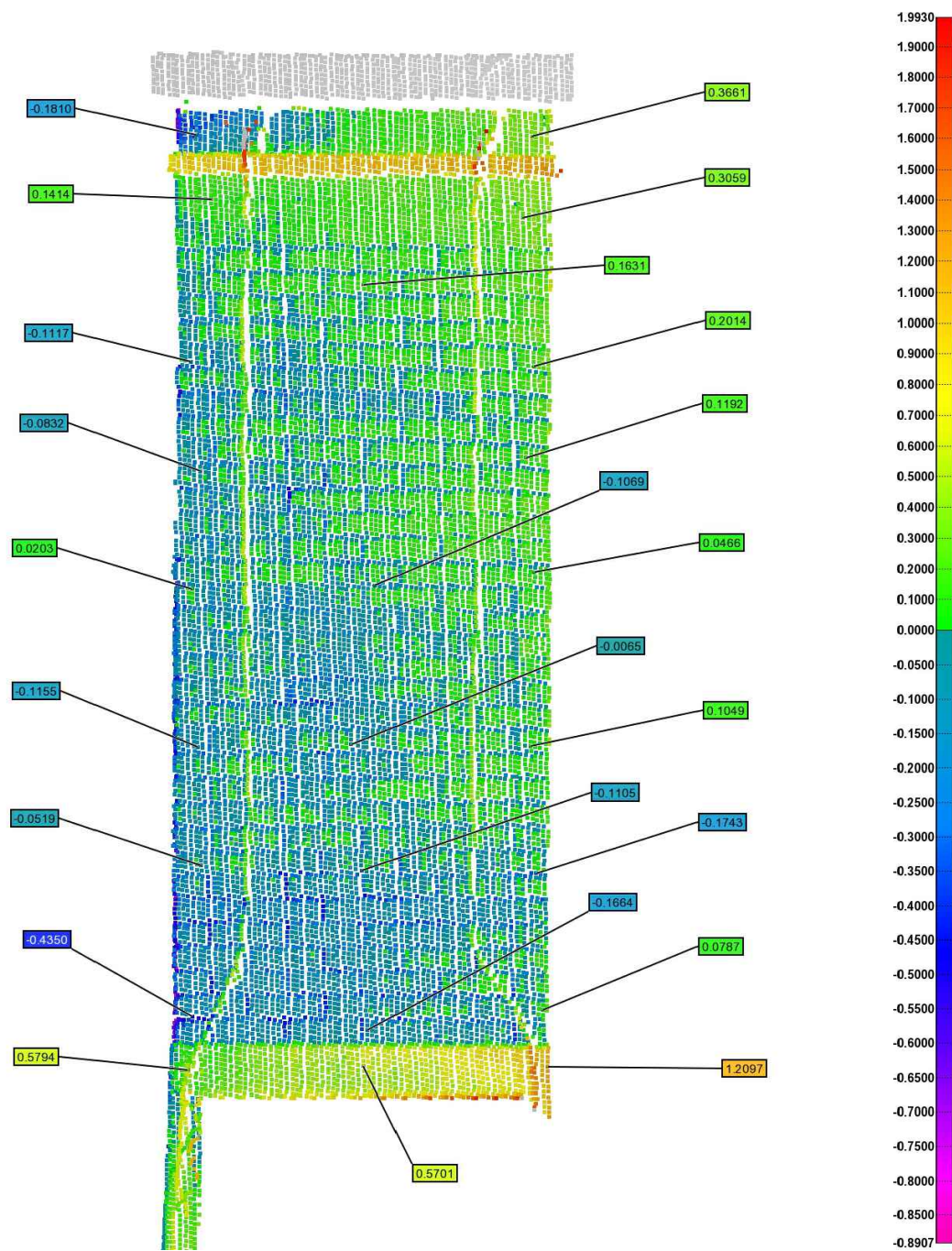
 John R. Davis Penna. Professional Engineer No. PE055730	A/E FIRM -- 24013 PRIME/ARCH: GWGO, INC. BALTIMORE, MD STRUCTURAL ENG: KEAST & HOOD PHILADELPHIA, PA	REVISIONS:			DESIGNED:	SUB SHEET NO. S2.2	TITLE OF SHEET CHIMNEY 2 REHABILITATE CONGRESS HALL MASONRY INDEPENDENCE NATIONAL HISTORICAL PARK	DRAWING NO.
		NO.	DATE	DESCRIPTION	BNF			391
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					JRD			SHEET
					DATE:			7 OF 15
			12/19/2024					



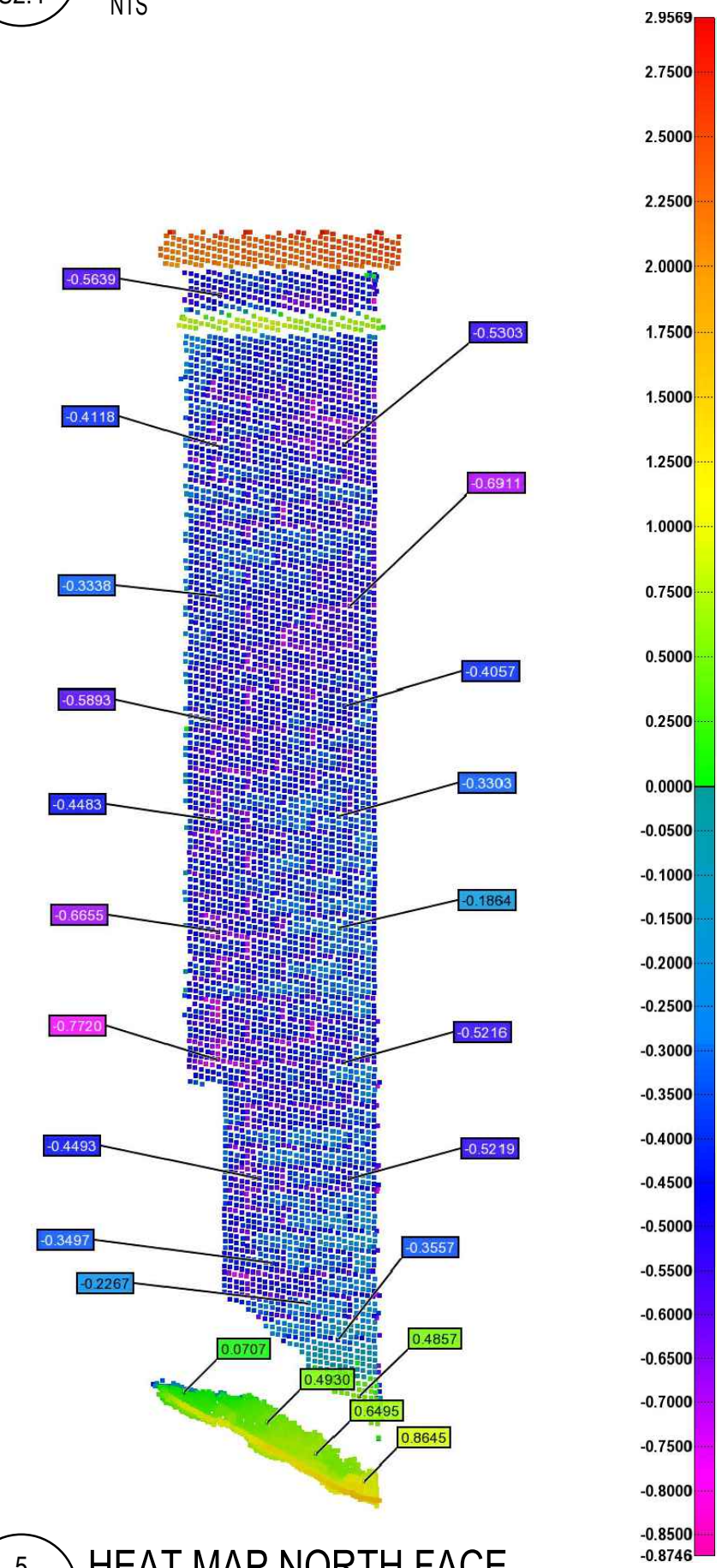
1 K&H SURVEY MEASUREMENTS
S2.4 NTS



4 HEAT MAP EAST FACE
S2.4 NTS

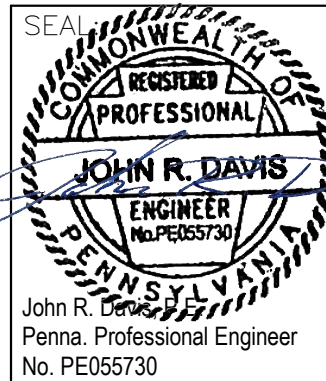


5 HEAT MAP NORTH FACE
S2.4 NTS



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PRIME/ARCH:
GWWO, INC.
BALTIMORE, MD
STRUCTURAL ENG:
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12/19/2024

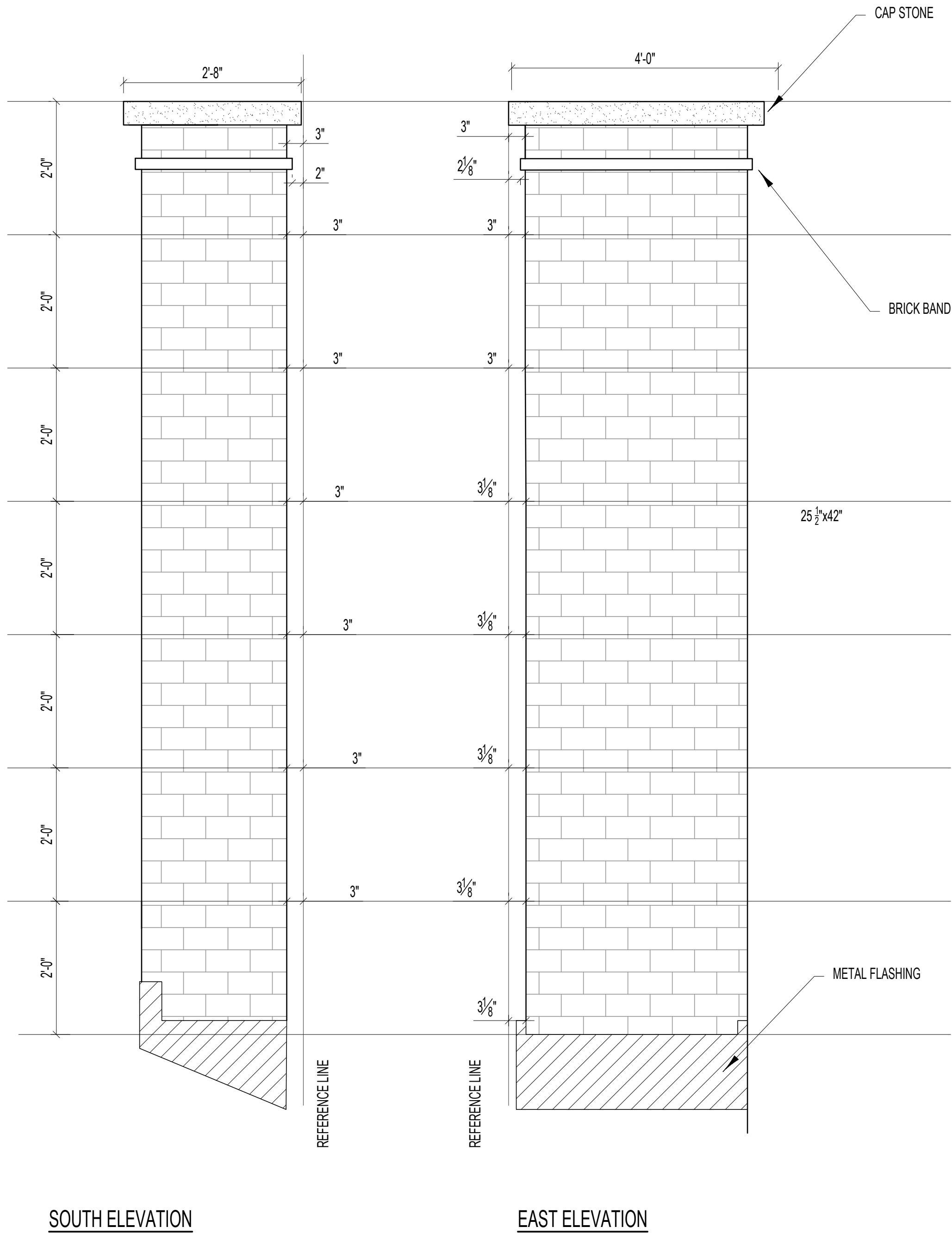
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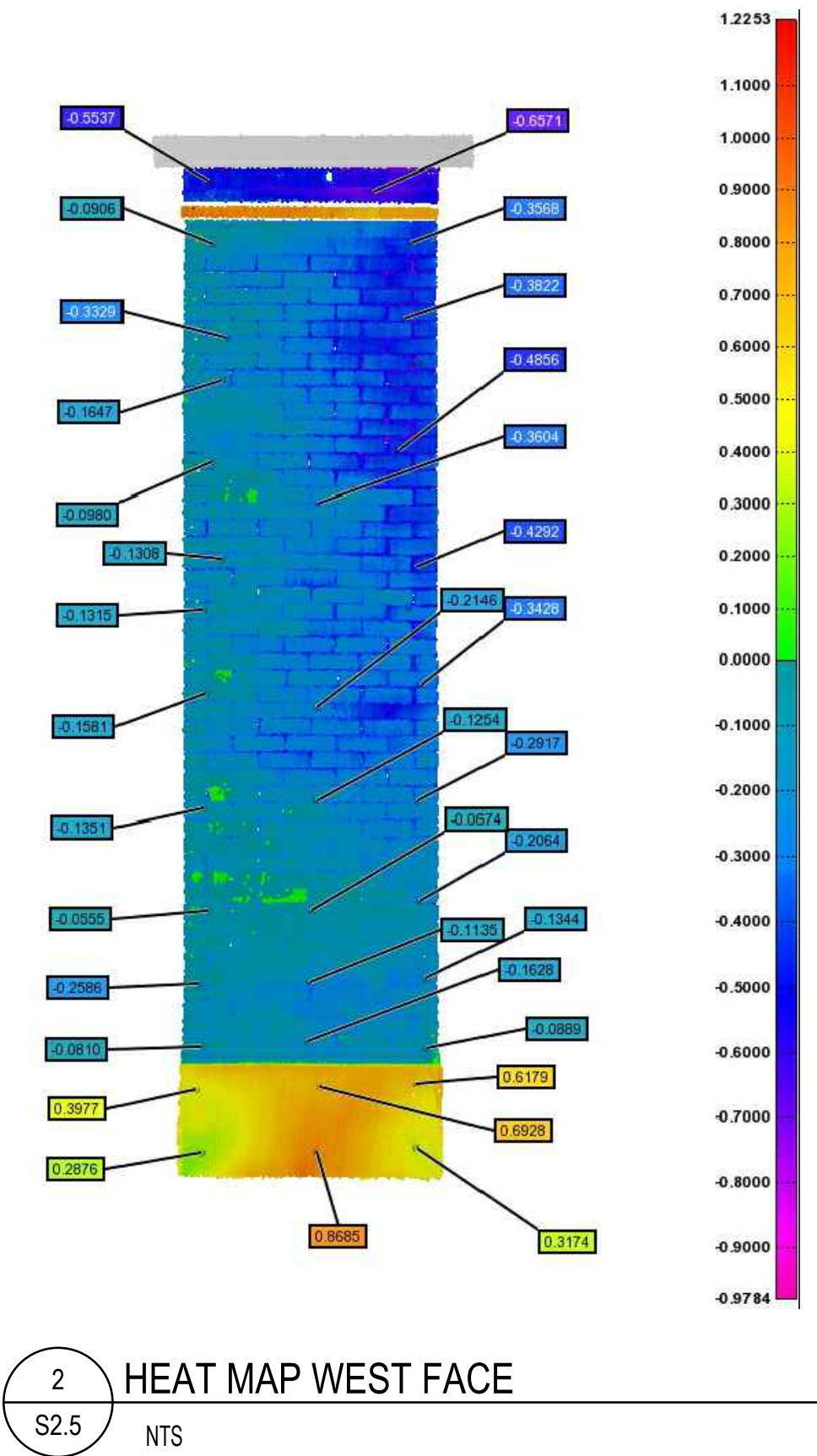
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CHIMNEY 4

REHABILITATE CONGRESS HALL MASONRY
INDEPENDENCE NATIONAL HISTORICAL PARK

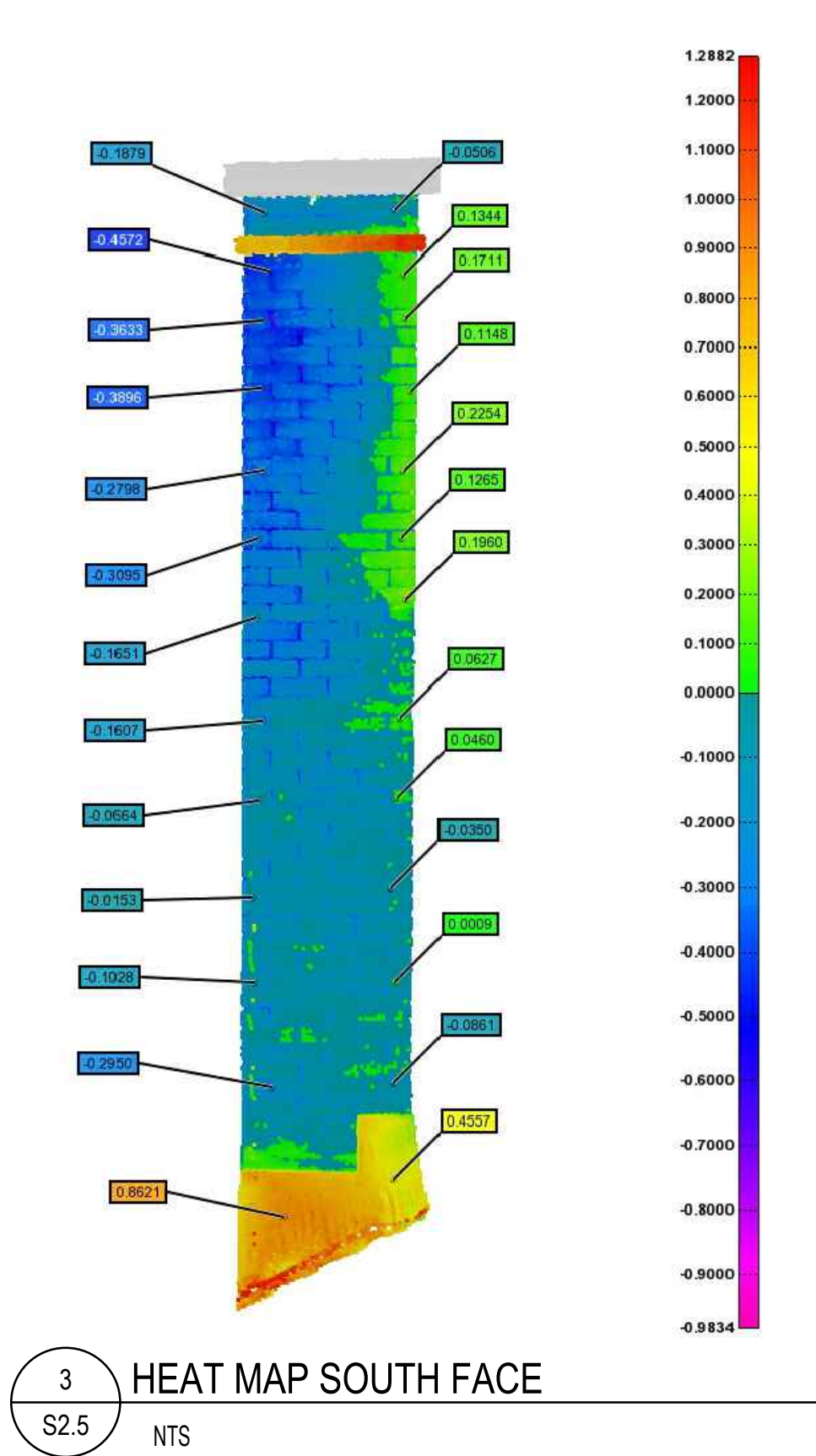
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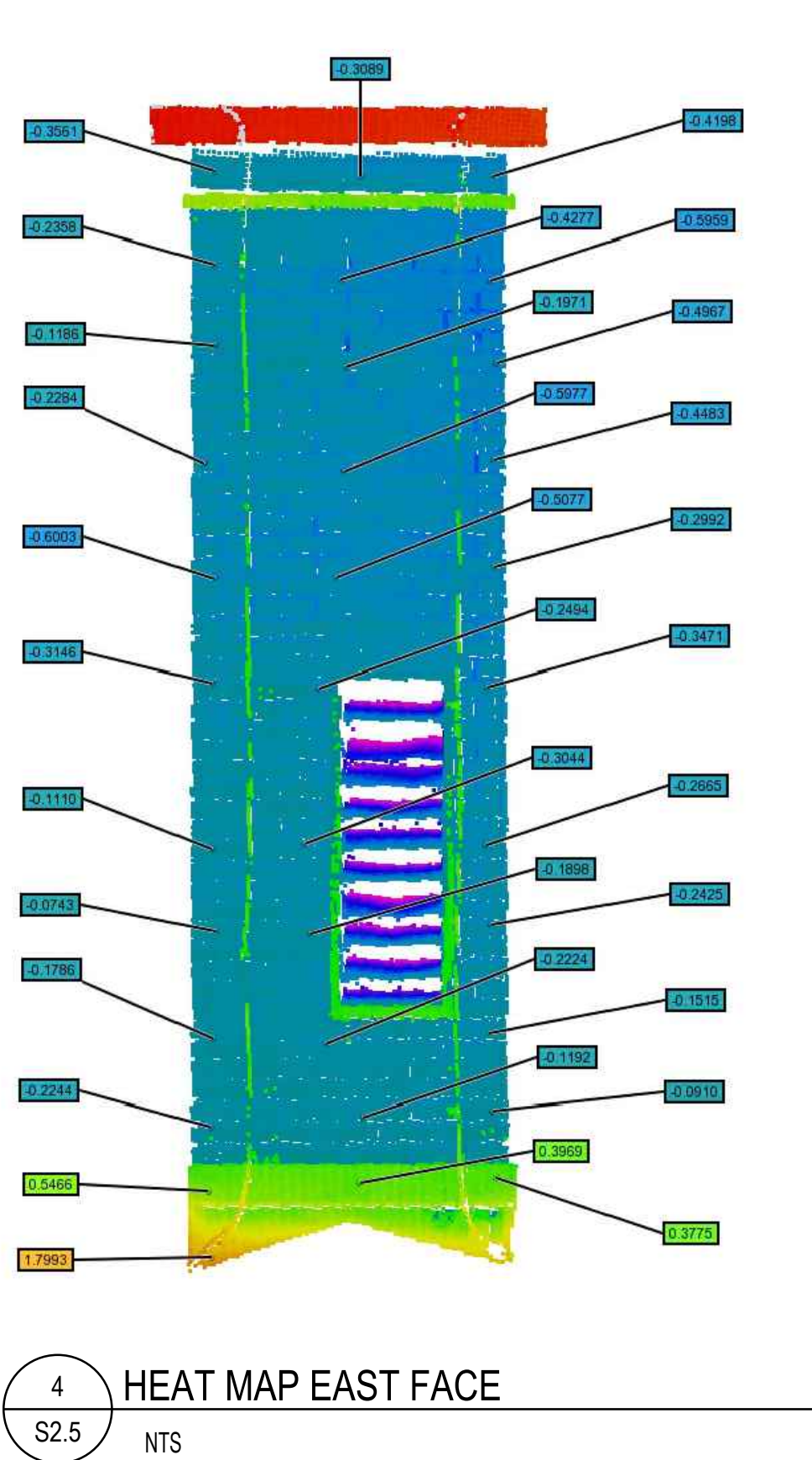
1 K&H SURVEY MEASUREMENTS
S2.5 NTS



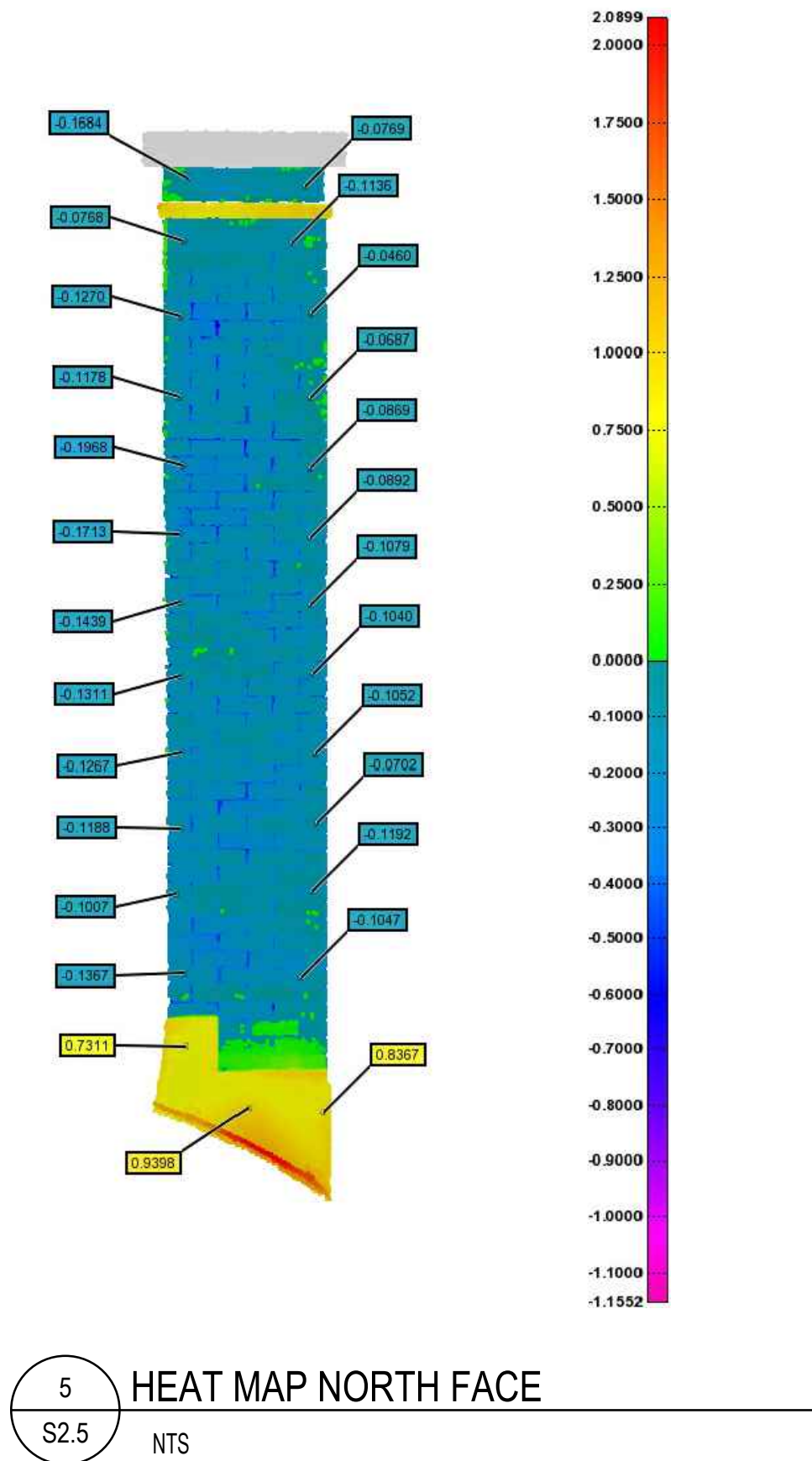
2 HEAT MAP WEST FACE
S2.5 NTS



3 HEAT MAP SOUTH FACE
S2.5 NTS



4 HEAT MAP EAST FACE
S2.5 NTS



5 HEAT MAP NORTH FACE
S2.5 NTS

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SEAL
JOHN R. DAVIS
REGISTERED PROFESSIONAL ENGINEER
No. PE055730
Penna. Professional Engineer

A/E FIRM - 24013
PRIME/ARCH:
GWMO, INC.
BALTIMORE, MD
STRUCTURAL ENG:
KEAST & HOOD
PHILADELPHIA, PA

REVISIONS:		
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TECH. REVIEW:
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DATE:
12/19/2024

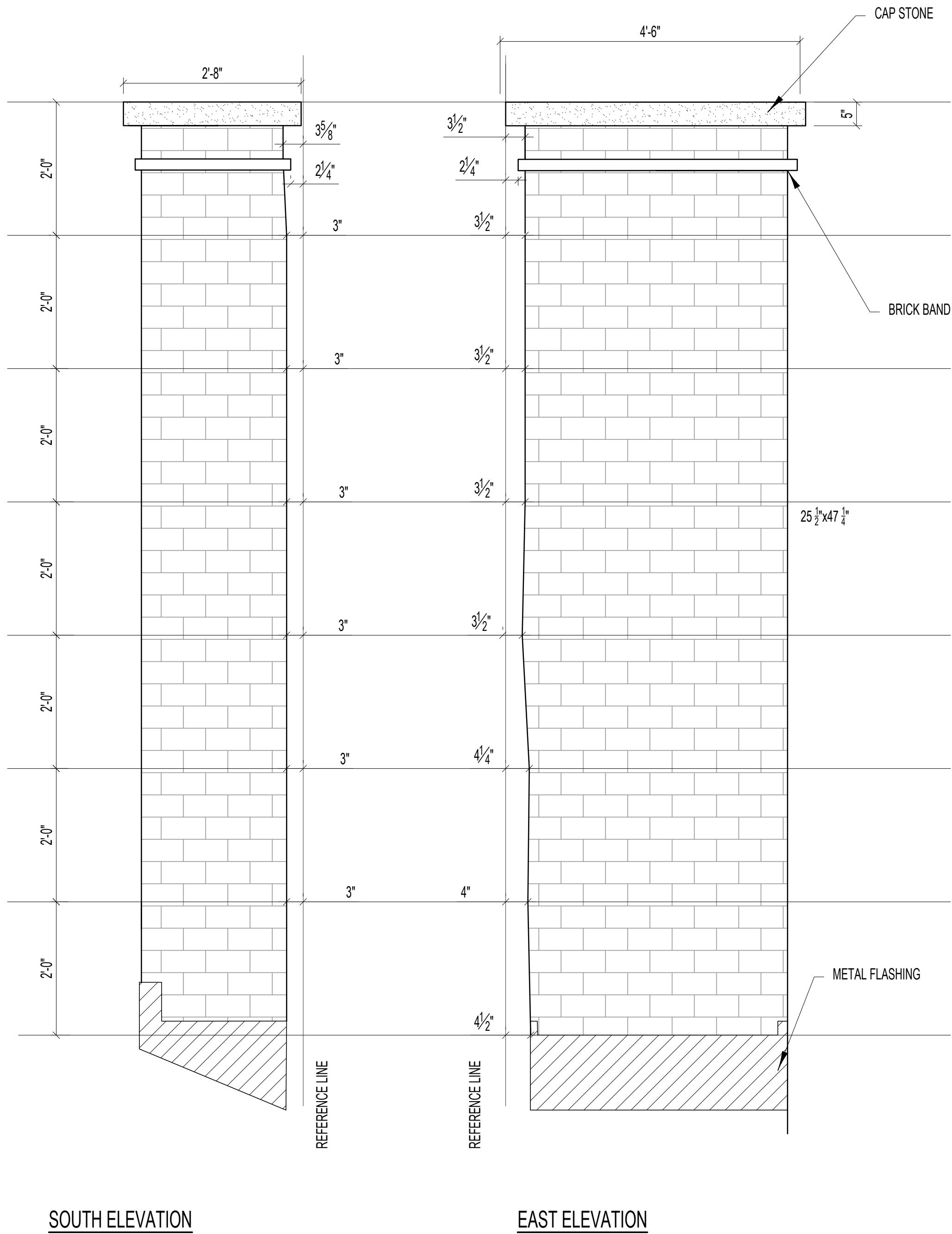
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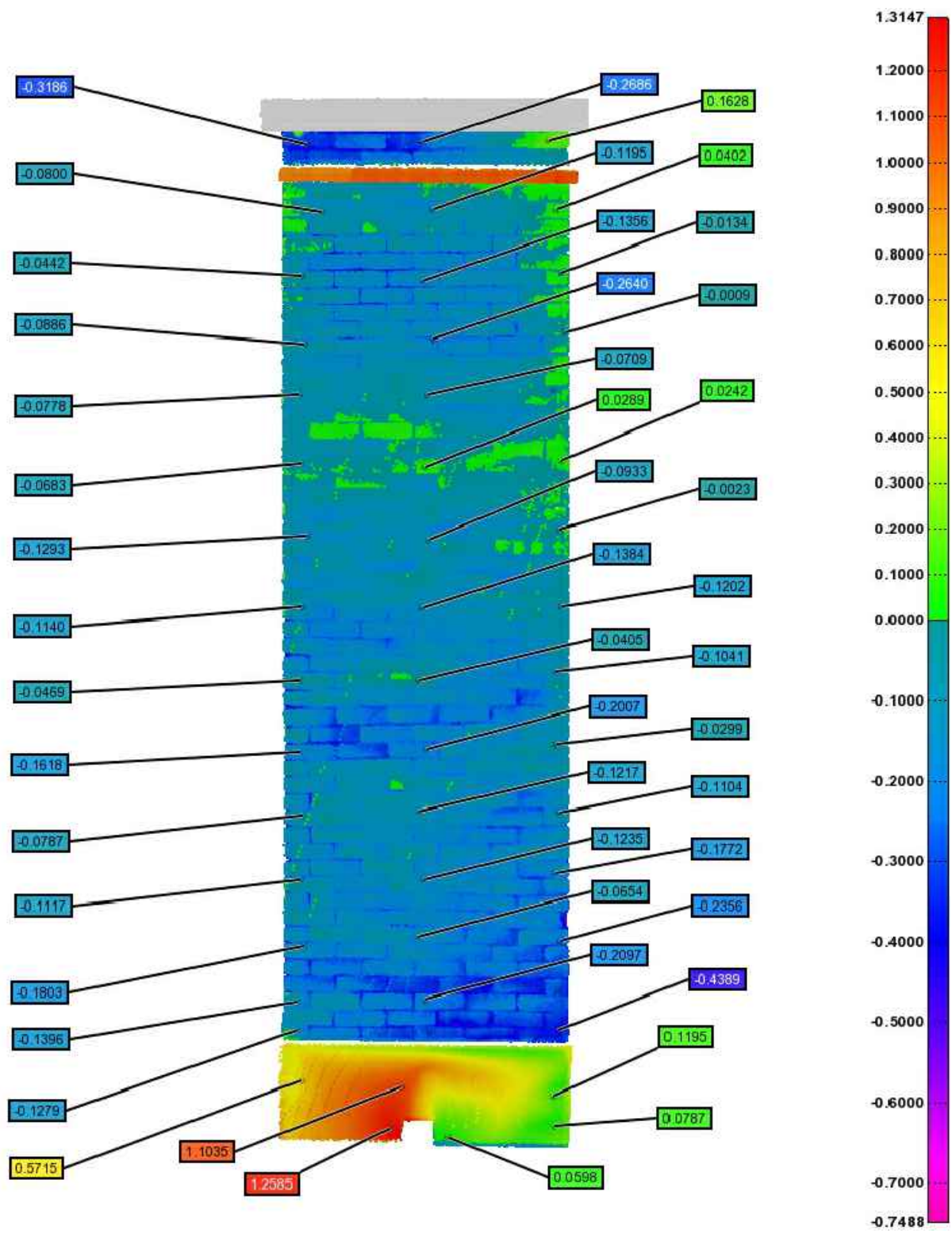
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CHIMNEY 5

REHABILITATE CONGRESS HALL MASONRY
INDEPENDENCE NATIONAL HISTORICAL PARK

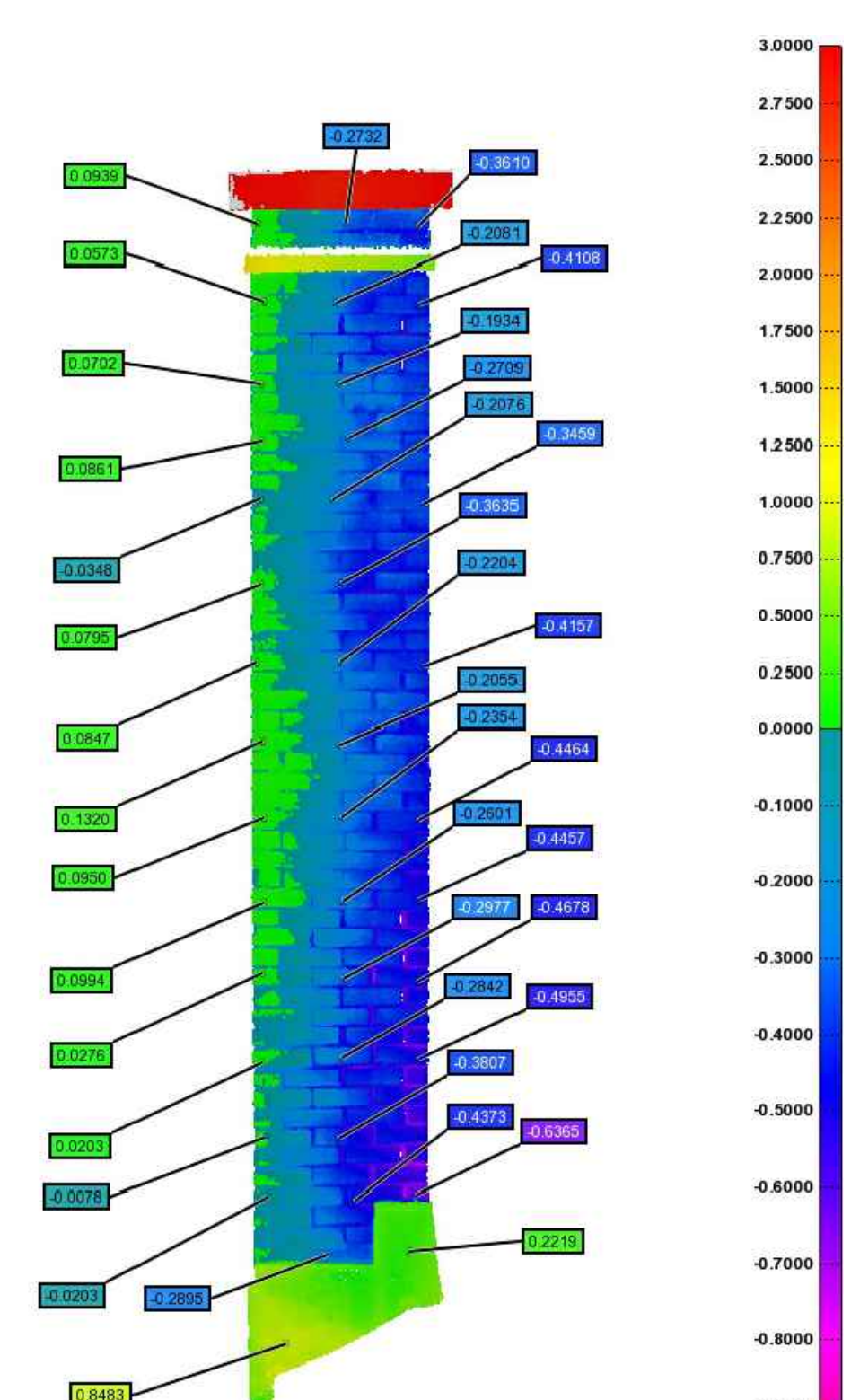
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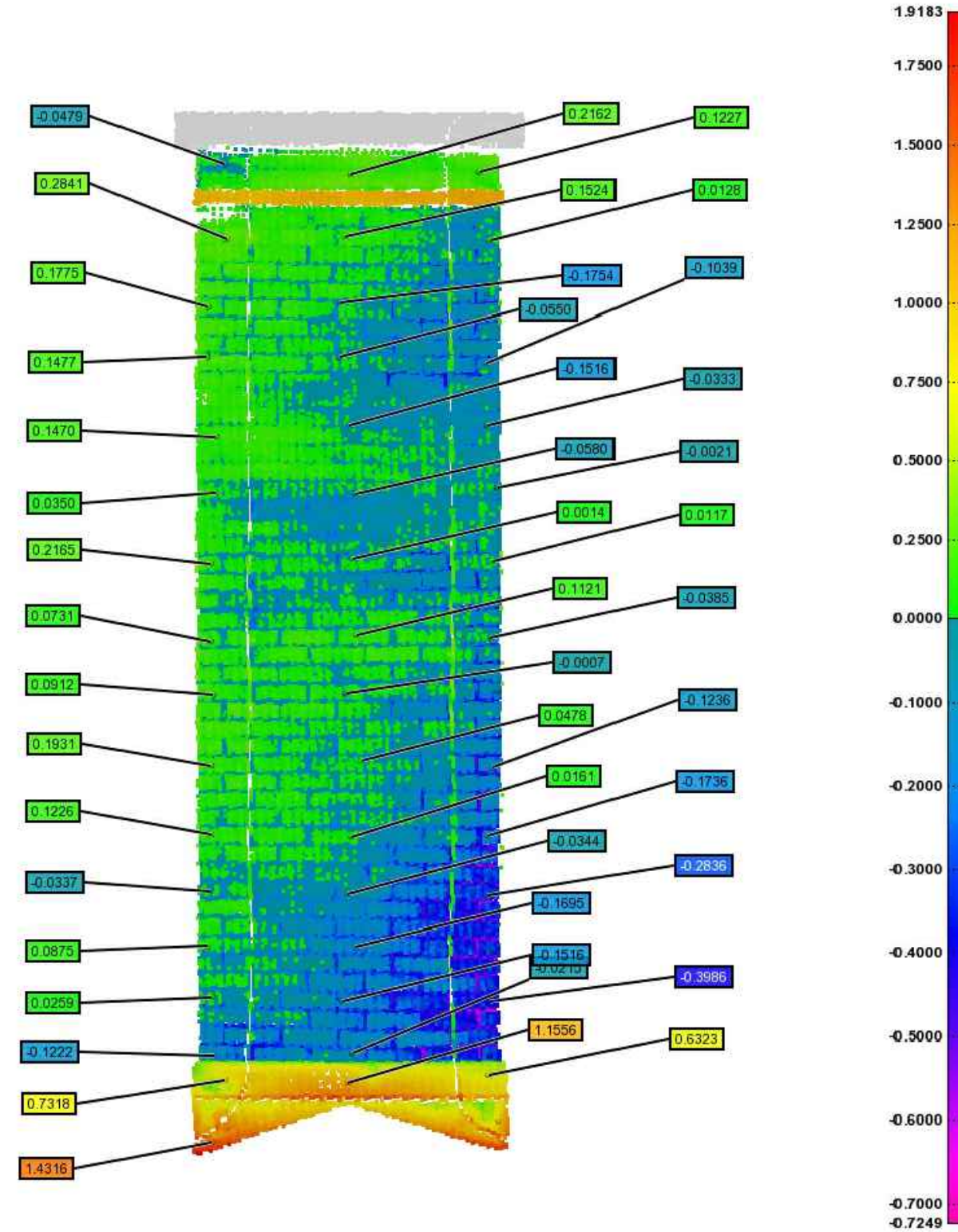
1 K&H SURVEY MEASUREMENTS
S2.6 NTS



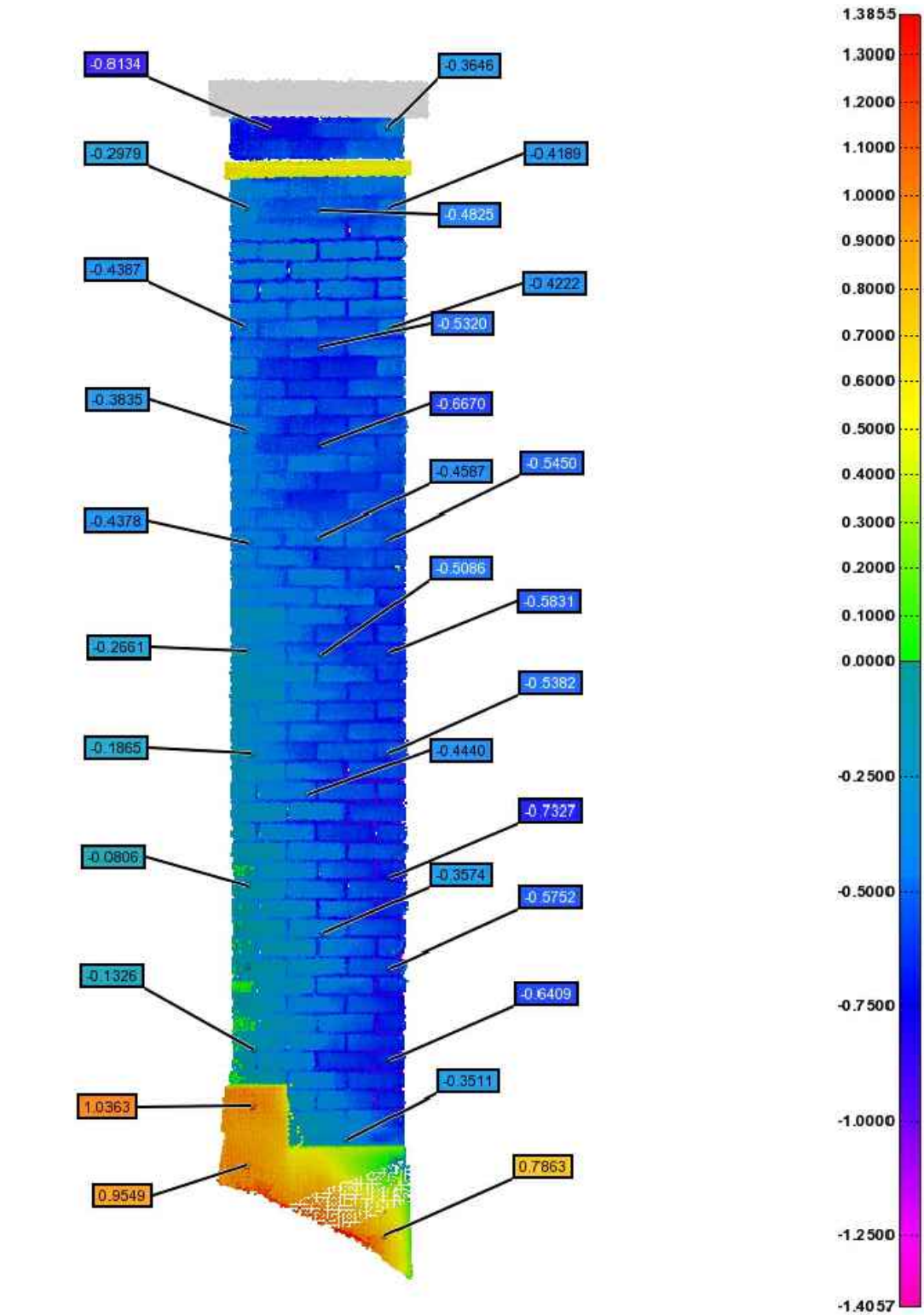
2 HEAT MAP WEST FACE
S2.6 NTS



3 HEAT MAP SOUTH FACE
S2.6 NTS



4 HEAT MAP EAST FACE
S2.6 NTS



5 HEAT MAP NORTH FACE
S2.6 NTS

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SEAL
PROFESSIONAL
ENGINEER
JOHN R. DAVIS
Penna. Professional Engineer
No. PE055730

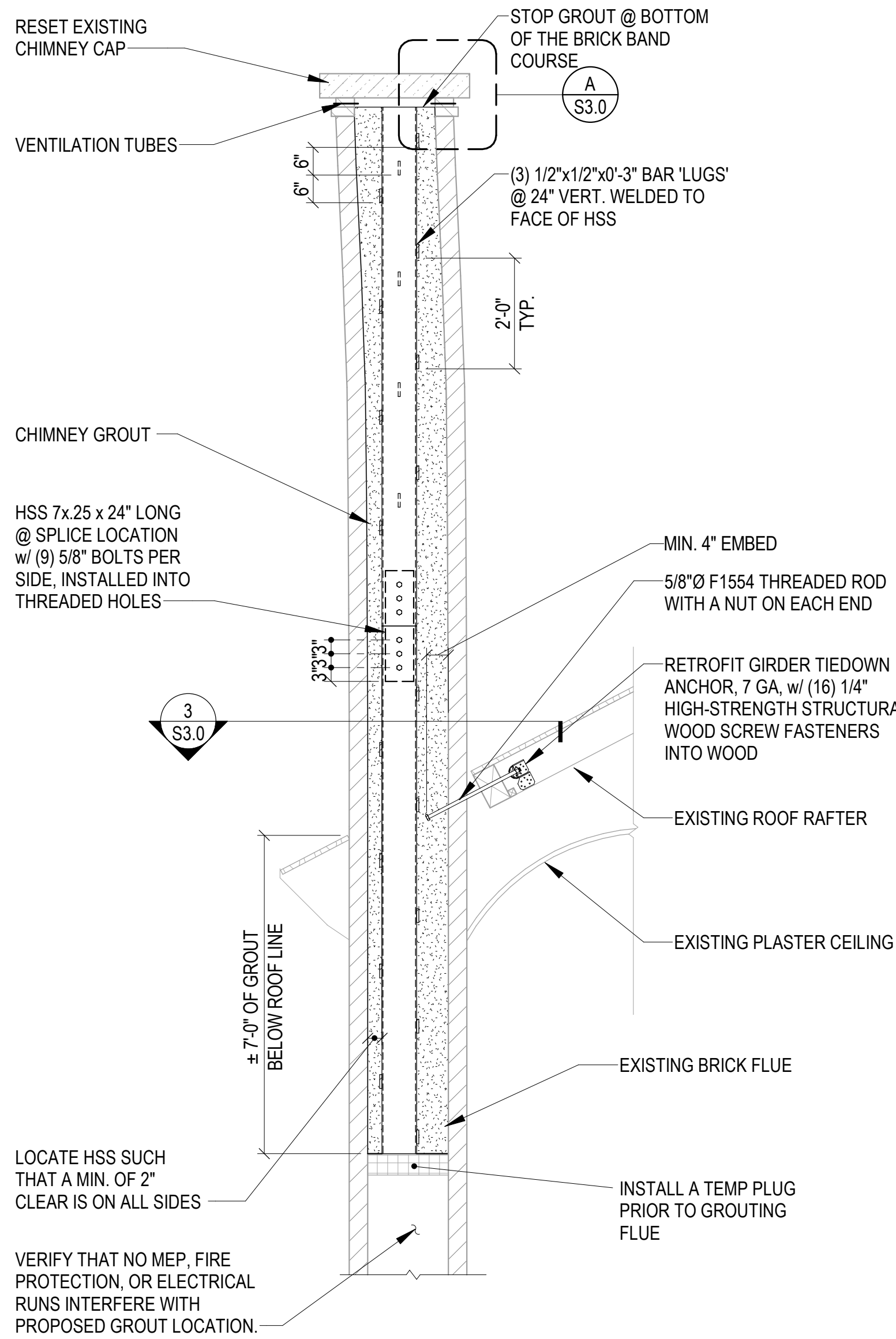
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STRUCTURAL ENG:
KEAST & HOOD
PHILADELPHIA, PA

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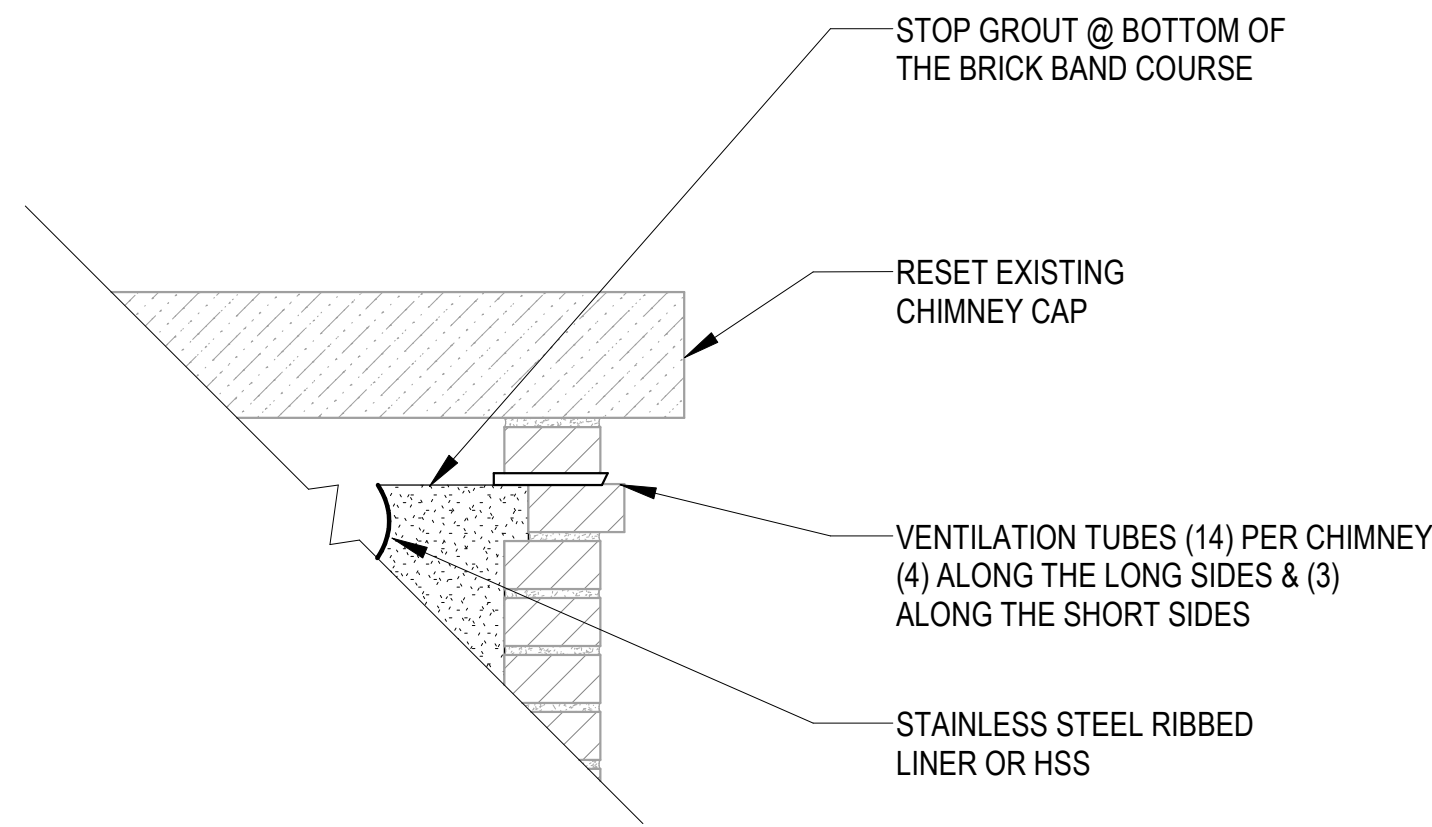
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TITLE OF SHEET
CHIMNEY 6
REHABILITATE CONGRESS HALL MASONRY
INDEPENDENCE NATIONAL HISTORICAL PARK

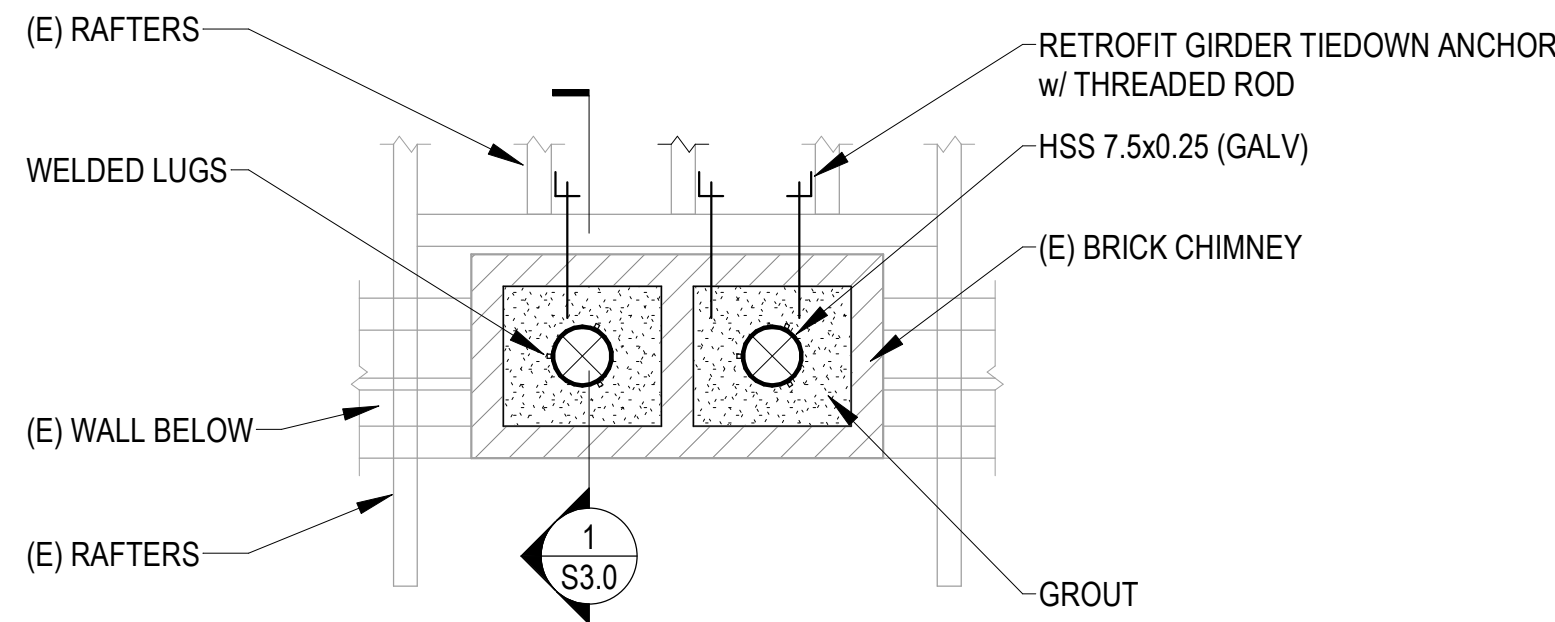
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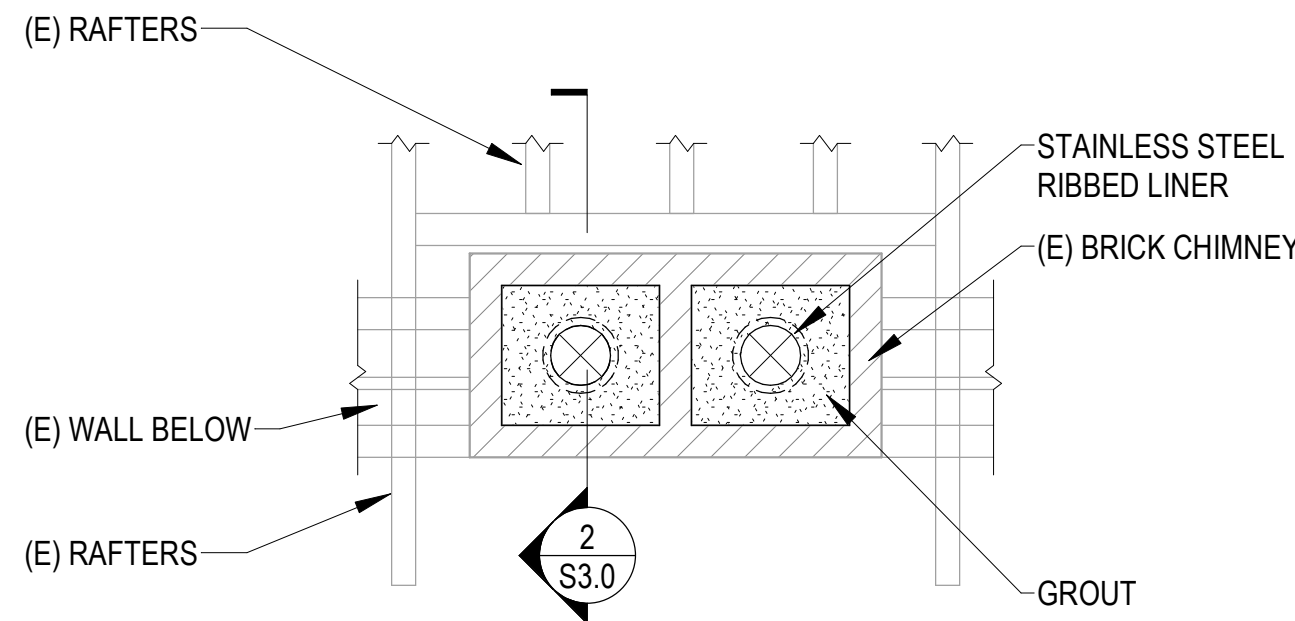
1 SECTION @ CHIMNEY 1 & CHIMNEY 2
S3.0 SCALE (A)



A
S3.0
DETAIL SECTION
SCALE (C)

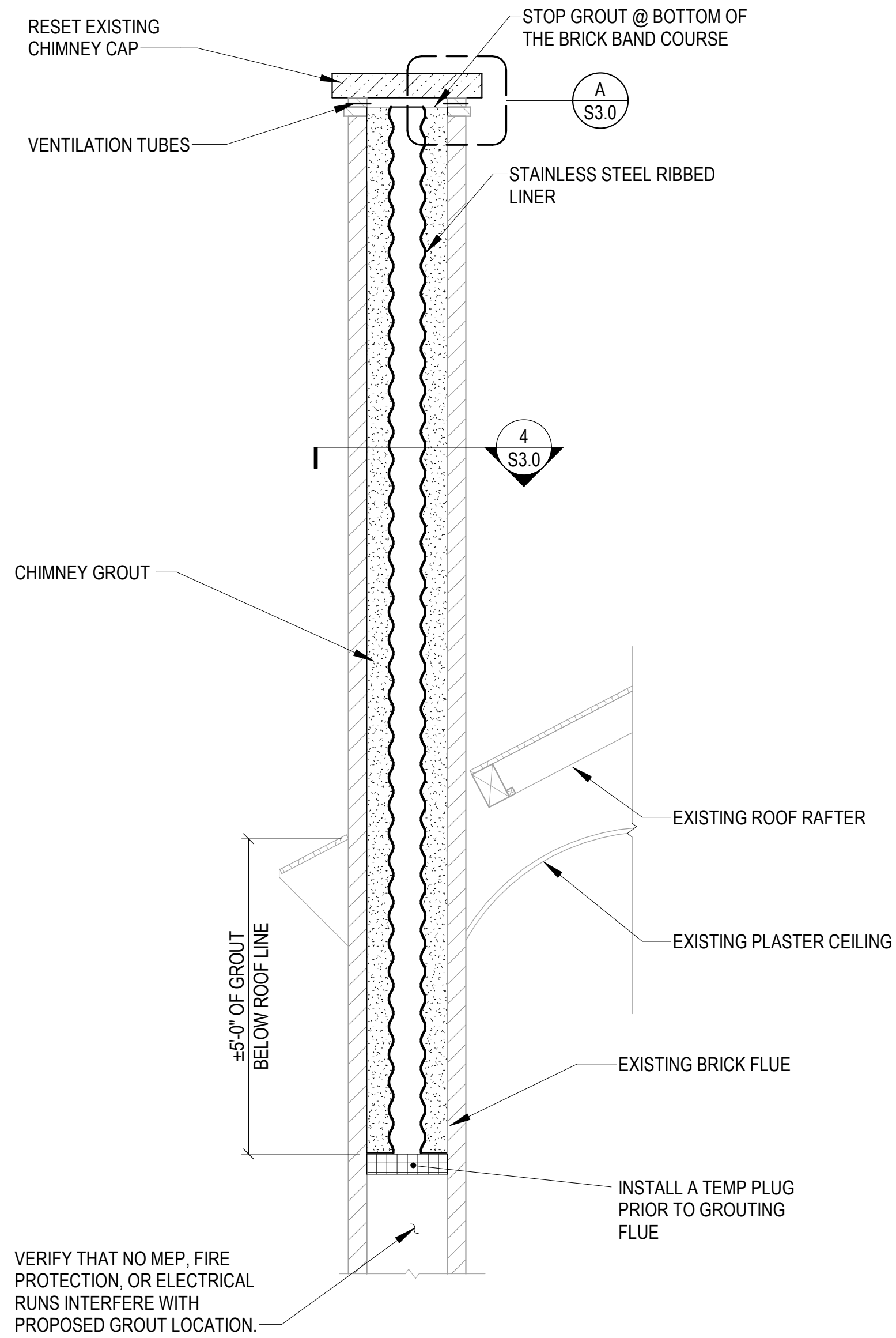


3 PARTIAL PLAN
S3.0 SCALE (A)

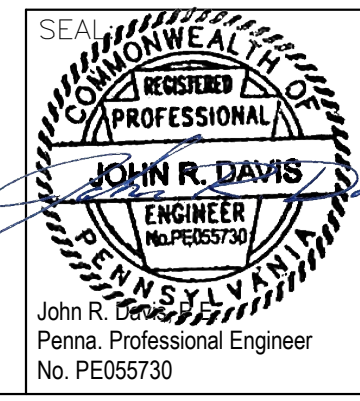
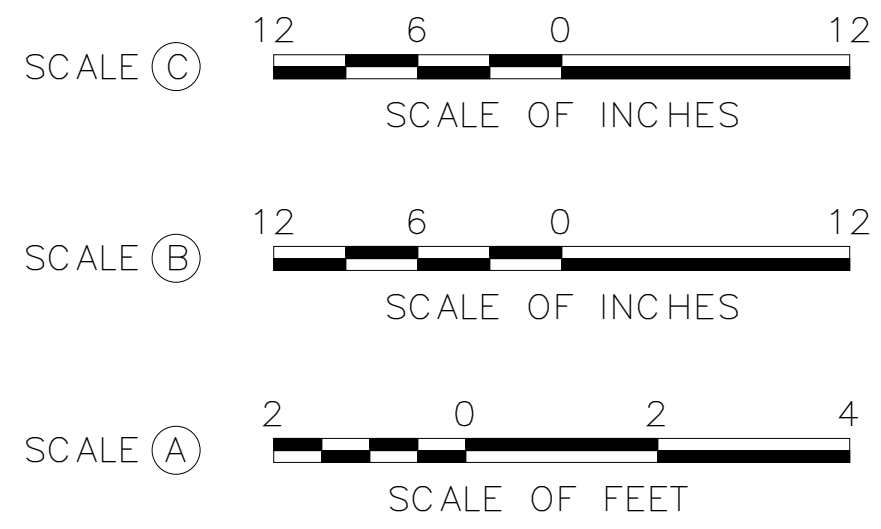


4 PARTIAL PLAN
S3.0 SCALE (A)

- Suggested Procedure for Chimney Lining:**
1. Provide top access to every flue at every chimney. Removal of stone cap is required. Salvage removed caps for reinstallation.
 2. Scope all flues at every chimney to check for breaches, blockages, and/or structural bearing. See SPEC for work plans and info.
 3. Perform necessary exterior masonry repairs to create continuous envelope including resetting or replacing bricks and repointing chimneys.
 4. Cleaning and Inspection of flue systems is to be performed by a CSIA certified chimney sweep. Testing And Approval Of Workers is required. Clean flue interior with rotary brush or pneumatic whip. Do not damage brick.
 5. Install temporary plug at predetermined point.
 6. Remove debris or residual soot deposits and loose mortar.
 7. Install ties to roof framing where indicated.
 8. Install stainless steel spring spacers to be wrapped around HSS or stainless liner to ensure adequate thickness of cast-in-place lining material.
 9. Wet flue interior to SSD prior to installing cast-in-place chimney lining material.
 10. Pour/pump cast-in-place light-weight cementitious chimney lining material that has been tested and listed to the UL1777 standard into annular space between chimney walls and flue lining system to stabilize and reinforce masonry chimney structure.
 11. Advance lining material equally in adjacent flues in no greater than 12" lifts so as to minimize uneven liner wall pressures. Consolidate and vibrate lining material throughout process.
 12. Upon completion of casting and after a minimum of 72 hours, complete work at chimney exterior including repointing.



2 SECTION @ CHIMNEYS 3, 4, 5, & 6
S3.0 SCALE (A)



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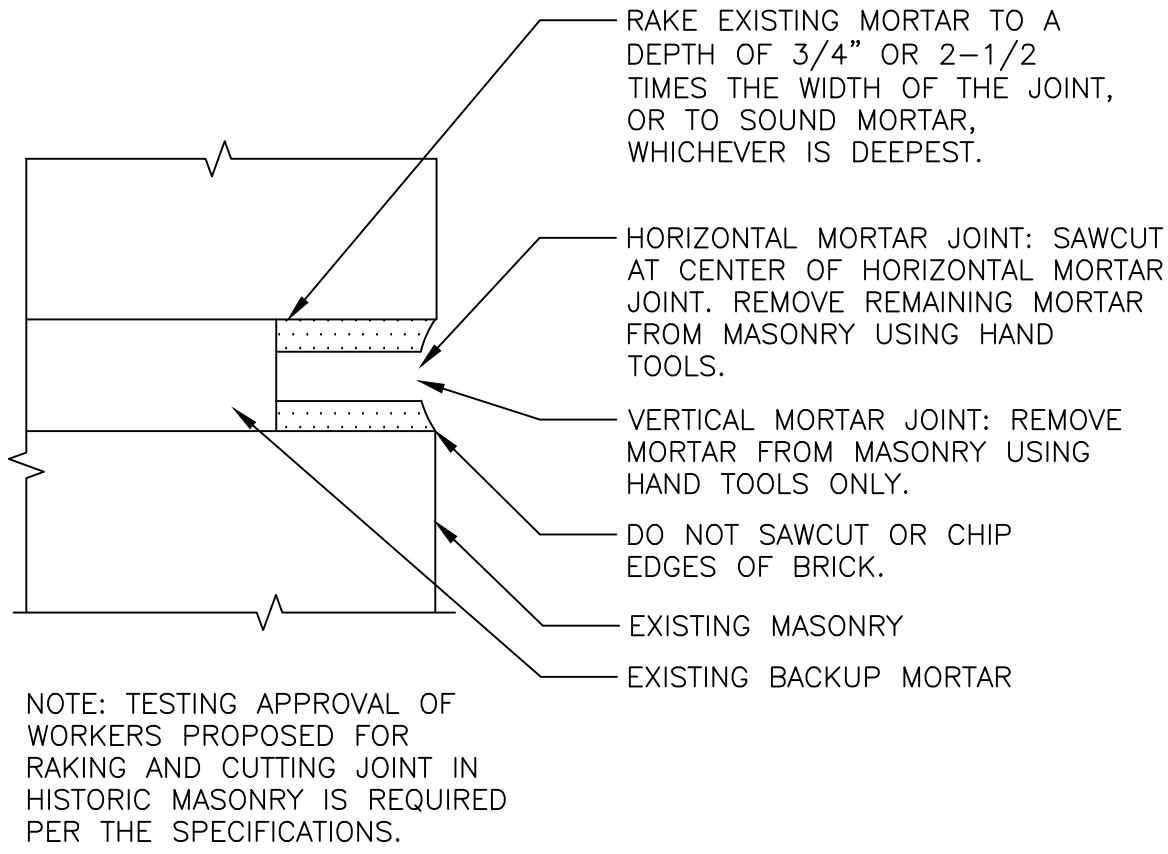
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TITLE OF SHEET
CHIMNEY REINFORCEMENT

REHABILITATE CONGRESS HALL MASONRY
INDEPENDENCE NATIONAL HISTORICAL PARK

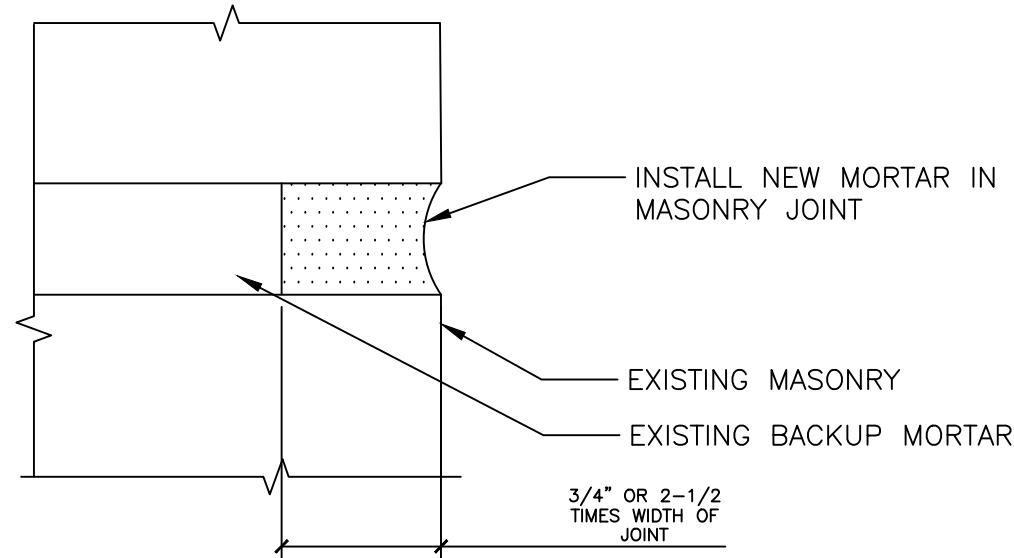
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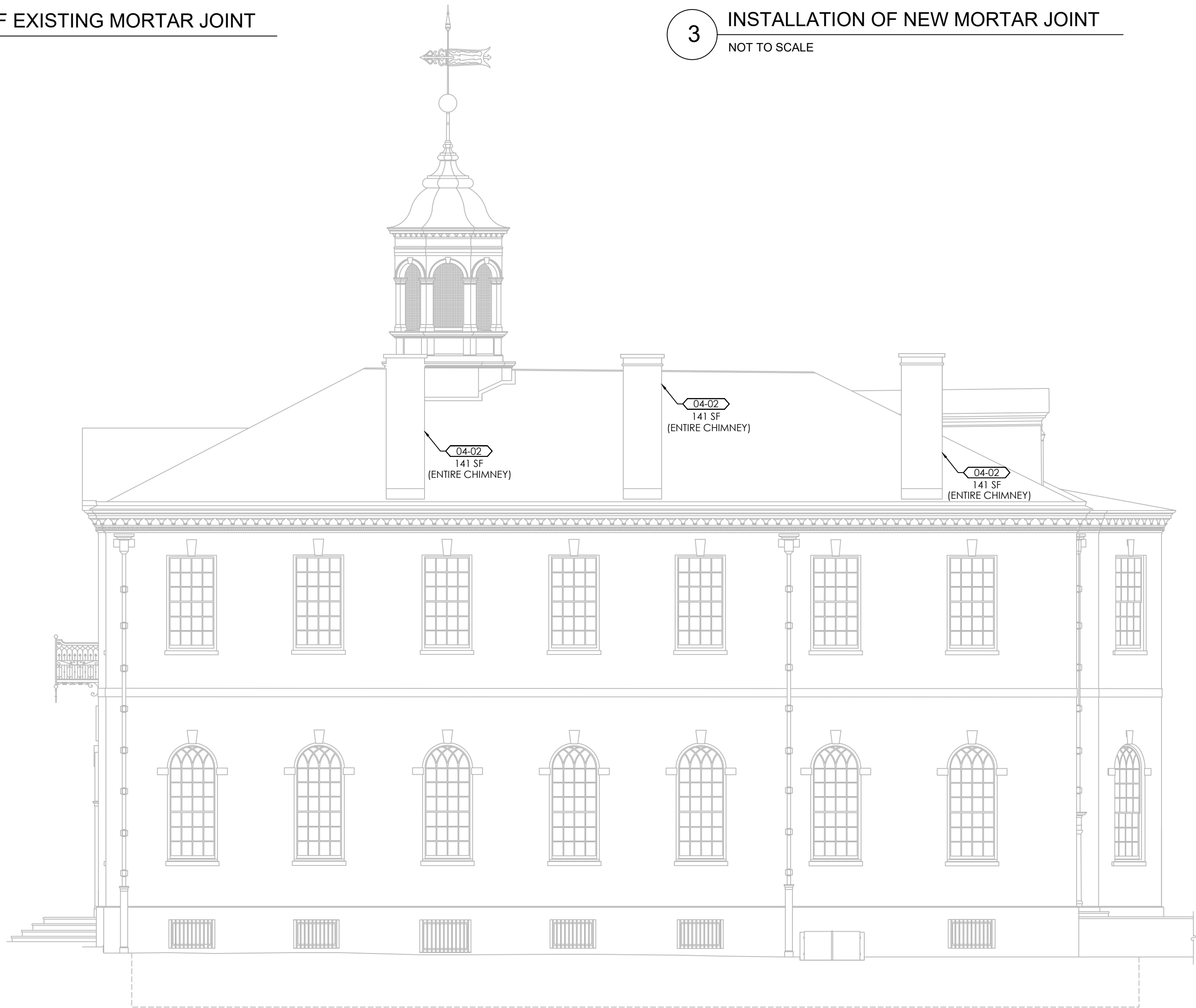


NOTE: TESTING APPROVAL OF WORKERS PROPOSED FOR RAKING AND CUTTING JOINT IN HISTORIC MASONRY IS REQUIRED PER THE SPECIFICATIONS.

2 REMOVAL OF EXISTING MORTAR JOINT
NOT TO SCALE



3 INSTALLATION OF NEW MORTAR JOINT
NOT TO SCALE



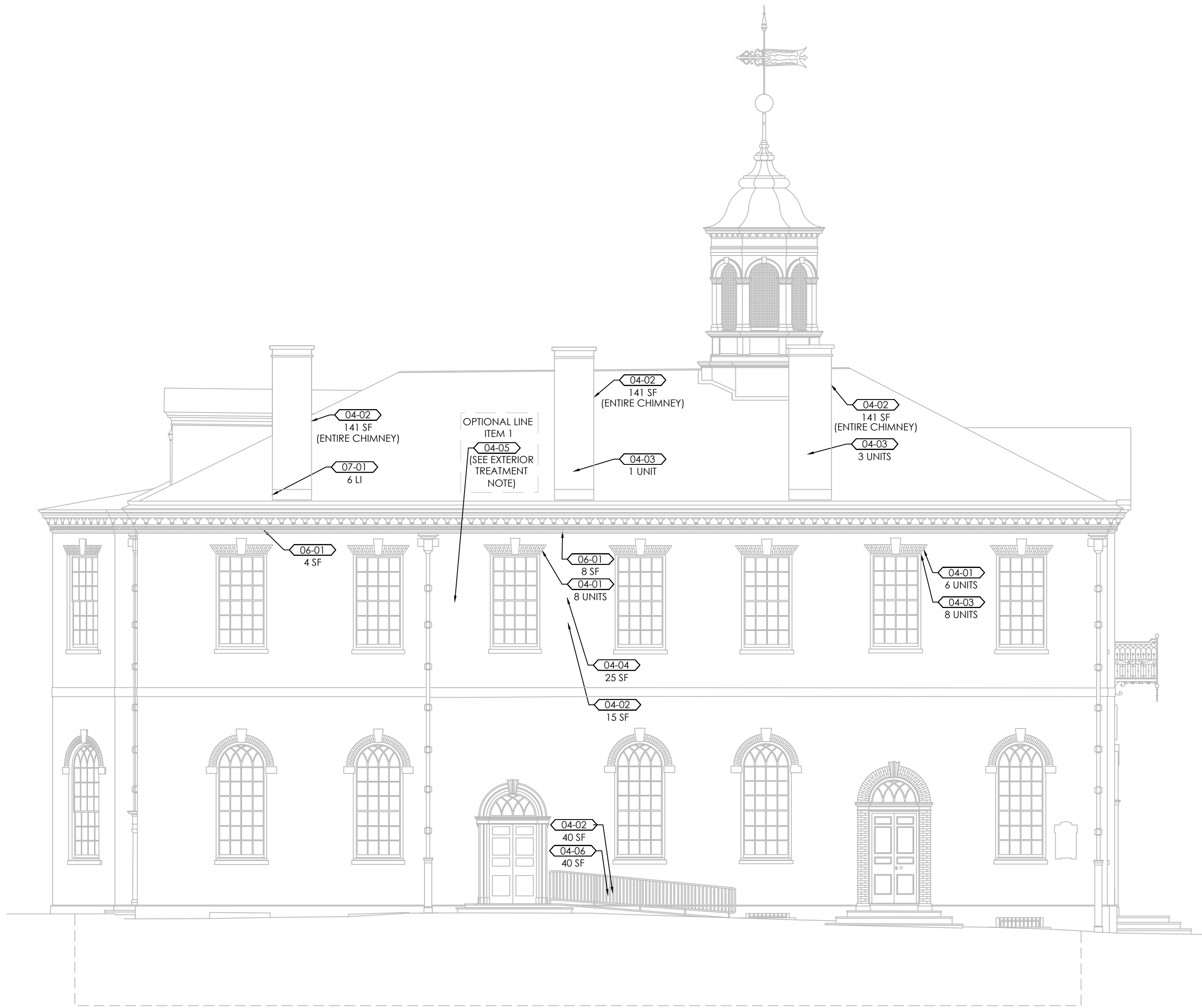
1 WEST ELEVATION
R3.1 SCALE (A)

EXTERIOR GENERAL NOTES		
1.	REMOVE 100% OF DEBRIS IN GUTTERS.	WORK ITEM 2
2.	PERFORM PROBE AT GUTTER ON EACH ELEVATION TO CONFIRM INSTALLED CONFIGURATION. REMOVE SHAKES AND REINSTALL OR REPLACE AS NECESSARY.	WORK ITEM 2
EXTERIOR TREATMENT NOTES		
04-01	REMOVE DAMAGED BRICKS FROM JACK ARCH. REBUILD PORTION OF JACK ARCH WITH NEW BRICKS TO MATCH ADJACENT.	OPTIONAL LINE ITEM 5
04-02	REMOVE EXISTING DETERIORATED MORTAR BACK TO SOUND MORTAR AND PROVIDE NEW MORTAR AT JOINTS IN BRICK MASONRY. SEE DETAILS 1/R3.1 AND 2/R3.1.	WORK ITEMS 1 & 2 OPTIONAL LINE ITEM 5
04-03	REMOVE PATCHED BRICKS AND REPLACE WITH NEW BRICKS TO MATCH ADJACENT.	WORK ITEM 2
04-04	REMOVE EFFLORESCENCE FROM BRICK SURFACES.	WORK ITEM 2
04-05	PROVIDE ADDITIONAL SPOT REPOINTING IN AREAS OF DAMAGED MORTAR IN BRICK MASONRY. REMOVE EXISTING MORTAR BACK TO SOUND MORTAR AND PROVIDE NEW MORTAR. ASSUME 5% REPOINTING OF TOTAL SF OF BUILDING OR 894 SF.	OPTIONAL LINE ITEM 5
04-06	REMOVE BIOLOGICAL GROWTH FROM BRICK SURFACES.	WORK ITEM 2
06-01	PROVIDE EPOXY REPAIR TO AREA OF DAMAGED WOOD. PREPARE AND PAINT ENTIRE ELEMENT SECTION AFTER REPAIR IS COMPLETE.	WORK ITEM 2
07-01	REMOVE EXCESS SOLDER, PREPARE, AND RESOLDER FAILED SOLDER JOINTS.	WORK ITEM 2



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	PRIME/ARCH: GWMO, INC. BALTIMORE, MD	NO.	DATE	DESCRIPTION	Designer			391	
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				12/19/2024					

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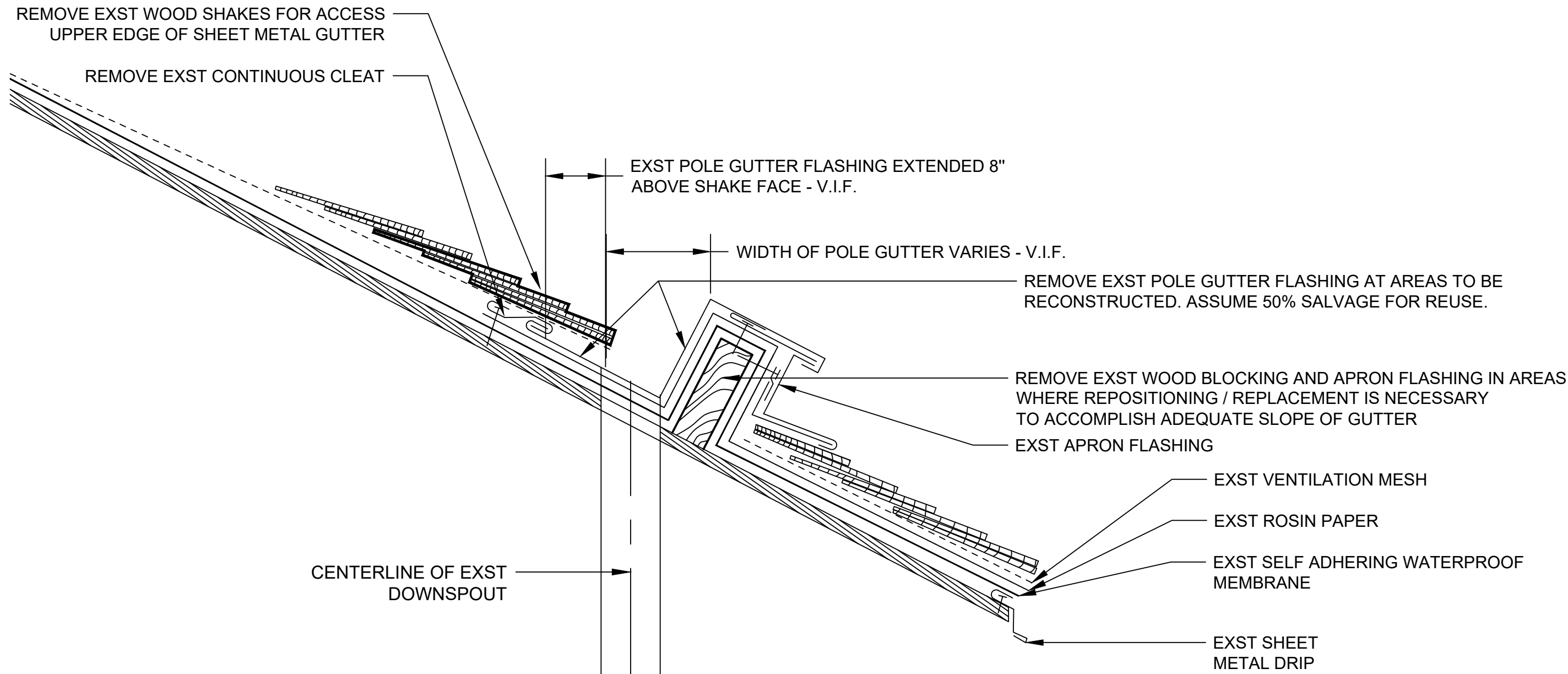
1 EAST ELEVATION
R3.2 SCALE A



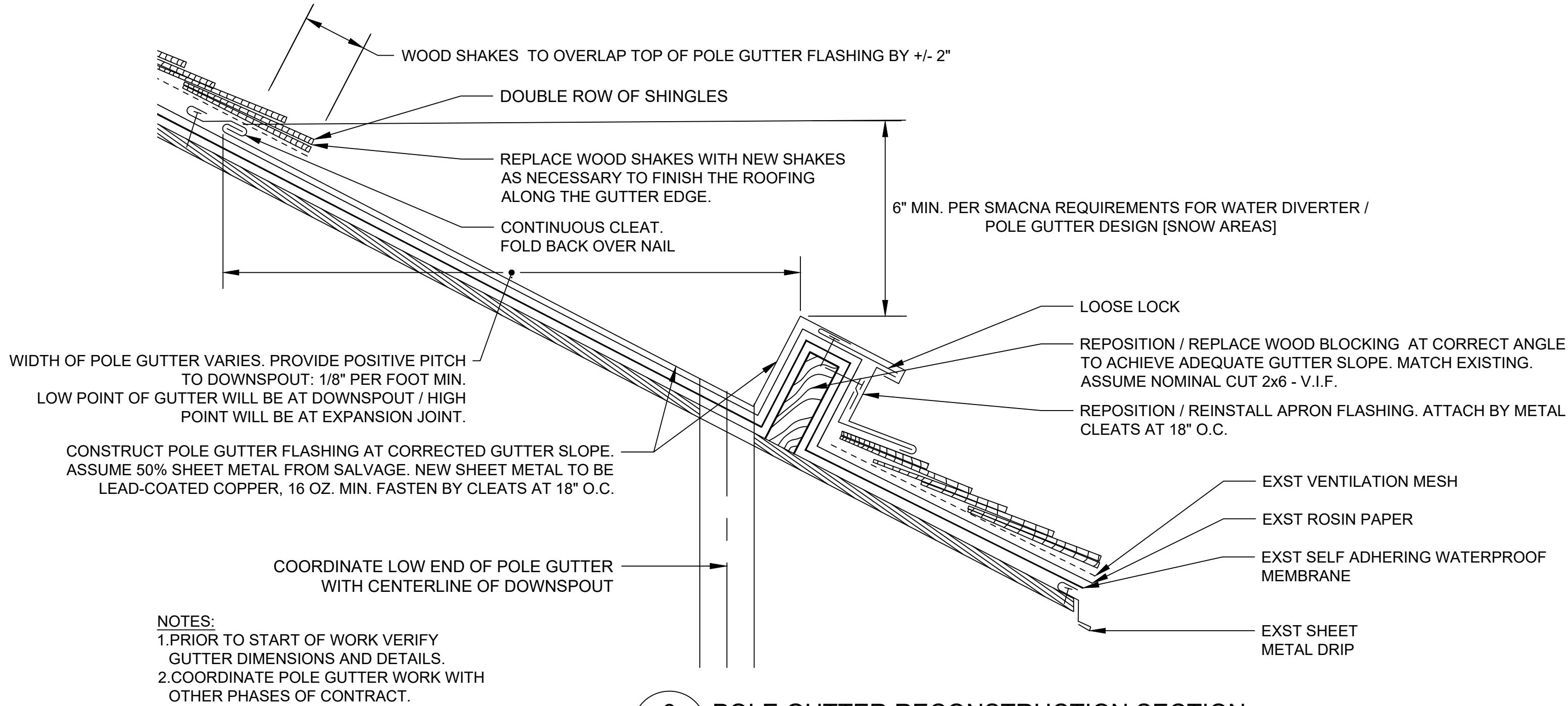
EXTERIOR GENERAL NOTES		
1.	REMOVE 100% OF DEBRIS IN GUTTERS.	WORK ITEM 2
2.	PERFORM PROBE AT GUTTER ON EACH ELEVATION TO CONFIRM INSTALLED CONFIGURATION. REMOVE SHAKES AND REINSTALL OR REPLACE AS NECESSARY.	WORK ITEM 2
EXTERIOR TREATMENT NOTES		
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04-03	REMOVE PATCHED BRICKS AND REPLACE WITH NEW BRICKS TO MATCH ADJACENT.	WORK ITEM 2
04-04	REMOVE EFFLORESCENCE FROM BRICK SURFACES.	WORK ITEM 2
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04-06	REMOVE BIOLOGICAL GROWTH FROM BRICK SURFACES.	WORK ITEM 2
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07-01	REMOVE EXCESS SOLDER, PREPARE, AND RESOLDER FAILED SOLDER JOINTS.	WORK ITEM 2

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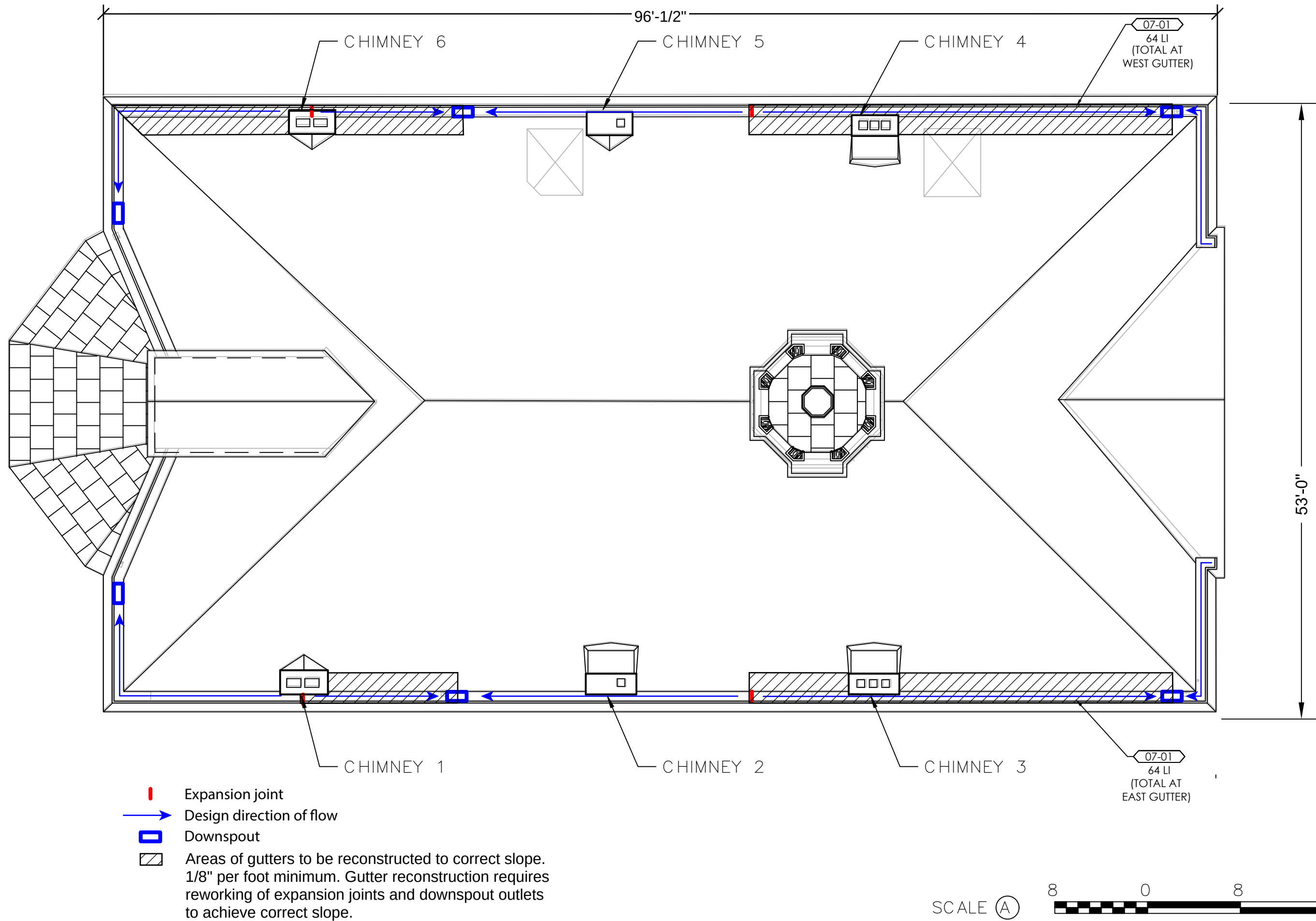


2
R3.3 SELECTIVE DEMOLITION AT EXISTING POLE GUTTER SECTION TO BE RECONSTRUCTED
SCALE: N.T.S.



3
R3.3 POLE GUTTER RECONSTRUCTION SECTION
SCALE: N.T.S.

EXTERIOR GENERAL NOTES		
1.	REMOVE 100% OF DEBRIS IN GUTTERS.	WORK ITEM 2
2.	PERFORM PROBE AT GUTTER ON EACH ELEVATION TO CONFIRM INSTALLED CONFIGURATION. REMOVE SHAKES AND REINSTALL OR REPLACE AS NECESSARY.	WORK ITEM 2
EXTERIOR TREATMENT NOTES		
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04-02	REMOVE EXISTING DETERIORATED MORTAR BACK TO SOUND MORTAR AND PROVIDE NEW MORTAR AT JOINTS IN BRICK MASONRY. SEE DETAILS 1/R3.1 AND 2/R3.1.	WORK ITEMS 1 & 2 OPTIONAL LINE ITEM 5
04-03	REMOVE PATCHED BRICKS AND REPLACE WITH NEW BRICKS TO MATCH ADJACENT.	WORK ITEM 2
04-04	REMOVE EFFLORESCENCE FROM BRICK SURFACES.	WORK ITEM 2
04-05	PROVIDE ADDITIONAL SPOT REPOINTING IN AREAS OF DAMAGED MORTAR IN BRICK MASONRY. REMOVE EXISTING MORTAR BACK TO SOUND MORTAR AND PROVIDE NEW MORTAR. ASSUME 5% REPOINTING OF TOTAL SF OF BUILDING OR 894 SF.	OPTIONAL LINE ITEM 5
04-06	REMOVE BIOLOGICAL GROWTH FROM BRICK SURFACES.	WORK ITEM 2
06-01	PROVIDE EPOXY REPAIR TO AREA OF DAMAGED WOOD. PREPARE AND PAINT ENTIRE ELEMENT SECTION AFTER REPAIR IS COMPLETE.	WORK ITEM 2
07-01	REMOVE EXCESS SOLDER, PREPARE, AND RESOLDER FAILED SOLDER JOINTS.	WORK ITEM 2



1
R3.3 ROOF PLAN
SCALE (A)

SEAL:	A/E FIRM -- 24013 PRIME/ARCH: GWGO, INC. BALTIMORE, MD	REVISIONS:			DESIGNED: Designer CADD: Author TECH. REVIEW: Checker DATE: 12/19/24	SUB SHEET NO. R3.3	TITLE OF SHEET ROOF PLAN SELECT EXTERIOR RESTORATION REHABILITATE CONGRESS HALL MASONRY INDEPENDENCE NATIONAL HISTORICAL PARK	DRAWING NO. 391 190326 PMIS/PKG. NO. 320146B SHEET 15 OF 15
		NO.	DATE	DESCRIPTION				